

POWERFACTORY

# PowerFactory 2024

## Technical Reference

### Common Result Variables for Terminals and Elements

Load Flow Calculation

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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January 9, 2024
PowerFactory 2024
Revision 1

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# 1 General Description

This document describes the common variables available for monitoring in *PowerFactory* for the terminals and for the single- and multiple-port elements (primary equipment). These are the parameters that can be selected to be displayed in the result boxes and in the flexible data page of the elements, which are not specific to a certain element.

Variables starting with capital letters are expressed in absolute values and variables starting with a lower case letters are expressed in per unit values.

## 1.1 Terminals

For the terminals (*ElmTerm*) only the set *Currents, Voltages and Powers* displays common result variables that can be monitored after a calculation. The identification name of a result variable contains the letter $m$ to denominate that it is a common monitoring variable (in opposite to $c$ which stands for calculation variable), a semicolon and the name of the variable. For example, the result variable *Voltage, Magnitude* has the following identification name $m$:$u$.

For the unbalanced representation, the phase result variables get a slightly different identification name where the name of the phase is added. For example: $m$:$u$:$A$.

## 1.2 Elements (single and multiple port)

For the single- and multiple-port elements (ex.: *ElmSym*, *ElmLne*, *ElmTr3*, etc.), there are two sets containing common result variables that can be monitored after a calculation:

- *Currents, Voltages and Powers*
- *Bus Results*

### 1.2.1 Currents, Voltages and Powers

The identification name of the variables available in this set is similar to the one used for the terminals with the difference that for the elements also the connection point name is added.

For example $m$:$i1$:$_LOCALBUS$ is the magnitude of the positive-sequence current of the connected element. If the result variable is shown for a certain type of an element, the real connection point name is used. For example $m$:$i1$:$bushv$ is the magnitude of the positive-sequence current flowing through the HV connection of a transformer and $m$:$i1$:$busi$ is the magnitude of the positive-sequence current flowing through the connection '$busi$' of a line.

For the unbalanced representation, the phase result variables get a slightly different identification name where the name of the phase is added. For example: $m$:$i1$:$_LOCALBUS$:$A$, $m$:$i1$:$bushv$: $A$, $m$:$I$:$bus1$:$A$.

The result variables available for the elements in p.u. values are based on the element (not on the terminal). Due to this, the same variable for a port element and a terminal may have a different value. For example, if the nominal voltage of an element differs from the nominal voltage of a terminal, the positive sequence voltage magnitude $m$:$u$ will have a different value compared to $m$:$u$ of a terminal.

### 1.2.2 Bus Results

The result variables available in the *Bus Results* set for the terminals are actually the variables from the connected terminal i.e. they are the same as the variables available in the *Currents, Voltages and Powers* set from the terminals.

There are two differences regarding the identification name:

- the letter '$n$' is used to denominate that this is a node (terminal) variable
- the connection-point name is also used

The result variables in p.u. values from this set are based on the terminal. If for a certain element $n{:}u1{:}bus1$ and $m{:}u1{:}bus1$ are displayed, the result will be different if the nominal voltage of the element differs from the nominal voltage of the terminal.

## 1.3 Grouping Elements

For the grouping elements (*ElmNet*, *ElmArea*, *ElmZone*), there is one set containing common result variables that can be monitored after a calculation:

- *Calculation Parameters*

# 2 Balanced Load Flow calculation

The balanced load flow calculations uses positive sequence network representation which is valid for balanced symmetrical networks. The resulting quantities are positive sequence quantities.

For DC terminals and elements, the variables displaying imaginary values are zero.

## 2.1 Result variables for terminals

As described in Section 1.2.2, the result variables from the *Bus Results* set from the port elements are equivalent to the result variables from the *Currents, Voltages and Currents* set for the terminals.

The result variables available for terminals after a balanced Load Flow calculation are presented in the following sub chapters.

### 2.1.1 Voltage-related variables for terminals

The voltage related variables for the terminals are all based on the positive-sequence, phase to ground complex voltage $\underline{u}$ resulting from the balanced load flow. The relationship between the absolute and per unit value voltage is $\underline{U} = \underline{u} \cdot U_{base}$ where the base voltage is $U_{base} = uknom/\sqrt{3}$ where $uknom$ is the nominal line-to-line voltage of the terminal.

Table 2.1: Voltage related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $u1$ | p.u. | Positive-Sequence Voltage, Magnitude |
| $u1r$ | p.u. | Positive-Sequence Voltage, Real-Part |
| $u1i$ | p.u. | Positive-Sequence Voltage, Imaginary-Part |
| $u$ | p.u. | Voltage, Magnitude |
| $ur$ | p.u. | Voltage, Real Part |
| $ui$ | p.u. | Voltage, Imaginary Part |
| $U$ | kV | Line-Ground Voltage, Magnitude |
| $Ul$ | kV | Line-Line Voltage, Magnitude |
| $phiu$ | deg | Voltage, Angle |
| $phiurel$ | deg | Voltage, Relative Angle |
| $u1pc$ | % | Voltage, Magnitude |
| $upc$ | % | Voltage, Magnitude |
| $du$ | % | Voltage Deviation |

The result variables from Table 2.1 are calculated as follows:

- $u1$ is obtained as the magnitude of the complex voltage $\underline{u}$ as:

$$u1 = |\underline{u}|$$

- $u1r$ is obtained from the complex voltage $\underline{u}$ as:

$$u1r = \Re(\underline{u})$$

- $u1i$ is obtained from the complex voltage $\underline{u}$ as:

$$u1i = \Im(\underline{u})$$

- $u$ is equal to the positive sequence voltage magnitude:

$$u = u1$$

- $ur$ is obtained as:

$$ur = u1r$$

- $ui$ is obtained as:

$$ui = u1i$$

- $U$ is obtained as:

$$U = u \cdot uknom/\sqrt{3}$$

  where $uknom$ is the nominal voltage of the terminal.

- $Ul$ is obtained as:

$$Ul = u \cdot uknom$$

- $phiu$ is obtained from the complex voltage $\underline{u}$ as:

$$phiu = \arctan\left(\frac{\Im(\underline{u})}{\Re(\underline{u})}\right) \cdot \frac{180}{\pi}$$

- $phiurel$ is obtained using the *Initial Angle of Bus Voltage* $phiini$ basic data parameter as:

$$phiurel = phiu - phiini \cdot \frac{180}{\pi}$$

- $u1pc$ is obtained as:

$$u1pc = u \cdot 100$$

- $upc$ is obtained as:

$$upc = u \cdot 100$$

- $du$ is obtained as:

$$du = upc - 100$$

### 2.1.2 Aggregated power variables for terminals

Table 2.2: Aggregated power variables for terminals

| Name | Unit | Description |
|---|---|---|
| $Pgen$ | MW | Generation, Active Power |
| $Qgen$ | Mvar | Generation, Reactive Power |
| $Pmot$ | MW | Motor Load, Active Power |
| $Qmot$ | Mvar | Motor Load, Reactive Power |
| $Pload$ | MW | General Load, Active Power |
| $Qload$ | Mvar | General Load, Reactive Power |
| $Pcomp$ | MW | Compensation (Losses) |
| $Qcomp$ | Mvar | Compensation |
| $Pnet$ | MW | External Networks, Active Power |
| $Qnet$ | Mvar | External Networks, Reactive Power |
| $Pflow$ | MW | Power Flow, Active Power |
| $Qflow$ | Mvar | Power Flow, Reactive Power |
| $Pout$ | MW | Outgoing Flow, Active Power |
| $Qout$ | Mvar | Outgoing Power, Reactive Power |
| $Sout$ | MVA | Outgoing Power, Apparent Power |
| $cosphiout$ | | Outgoing Power, Power Factor |
| $Pbalance$ | MW | Active Power Balance (=0) |
| $Qbalance$ | Mvar | Reactive Power Balance (=0) |

The result variables from Table 2.2 are calculated as follows:

- $Pgen$ is the active power sum of all synchronous, asynchronous and static generators connected to the terminal. If generation is defined in the MV Load (*ElmLodmv*), the generated active power is also added to this sum.
- $Qgen$ is the reactive power sum of all synchronous, asynchronous and static generators connected to the terminal. If generation is defined in the MV Load (*ElmLodmv*), the generated reactive power is added to this sum.
- $Pmot$ is the active power sum of all synchronous and asynchronous motors connected to the terminal.
- $Qmot$ is the reactive power sum of all synchronous and asynchronous motors connected to the terminal.
- $Pload$ is the active power sum of all loads connected to the terminal.
- $Qload$ is the reactive power sum of all loads connected to the terminal.
- $Pcomp$ is the active power sum of all shunts and filters connected to the terminal.
- $Qcomp$ is the reactive power sum of all shunts and filters connected to the terminal.
- $Pnet$ is the active power sum of all external network elements connected to the terminal.
- $Qnet$ is the reactive power sum of all external network elements connected to the terminal.
- $Pflow$ is the sum of all active power flowing out of the terminal.
- $Qflow$ is the sum of all reactive power flowing out of the terminal.
- $Pout$ is the sum of all active power flowing out of the terminal.

- $Qout$ is the sum of reactive powers of all elements whose active power is flowing out of the terminal ($\sum Q$ if $P > 0$).
- $Sout$ is calculated as:

$$Sout = \sqrt{Pout^2 + Qout^2}$$

- $cosphiout$ is obtained as:

$$cosphiout = \arctan\left(\frac{Qout}{Pout}\right)$$

- $Pbalance$ is the sum of all active power at the terminal. This sum is always zero or near zero and is dependent on the maximum acceptable load flow error setting.
- $Qbalance$ is the sum of all reactive power at the terminal. This sum is always zero or near zero and is dependent on the maximum acceptable load flow error setting.

### 2.1.3 Low-voltage analysis related variables for terminals

Table 2.3: Low-voltage analysis related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $umin$ | p.u. | Minimum Voltage |
| $Umin$ | kV | Minimum Voltage (Line to Neutral) |
| $dumax$ | % | Maximum Voltage Drop along Feeder |
| $dUmax$ | kV | Maximum Voltage Drop along Feeder |
| $dUlmax$ | kV | Maximum Voltage Drop along Feeder (Line-Line) |
| $Ulmin$ | kV | Minimum Voltage (Line to Line) |

The result variables from Table 2.3 are calculated from a low-voltage load flow analysis where coincidence of low-voltage loads is considered.

### 2.1.4 Feeder losses variables for terminals

Table 2.4: Load flow, balanced, feeder losses variables (terminals)

| Name | Unit | Description |
|---|---|---|
| $LossPdown$ | MW | Losses, downstream |
| $LossQdown$ | Mvar | Losses, downstream (Reactive Power) |
| $LossPdownload$ | MW | Load losses, downstream |
| $LossQdownload$ | Mvar | Load losses, downstream |
| $LossPdownnoload$ | MW | No load losses, downstream |
| $LossQdownnoload$ | Mvar | No load losses, downstream |
| $du_feed$ | % | Voltage difference relative to feeder begin |

The result variables from Table 2.4 are calculated as follows:

- $LossPdown$ is the sum of active power load and no-load losses of all elements downward the feeder.

- $LossQdown$ is the sum of reactive power load and no-load losses of all elements downward the feeder.
- $LossPdownload$ is the sum of active power load losses of all elements downward the feeder.
- $LossQdownload$ is the sum of reactive power load losses of all elements downward the feeder.
- $LossPdownnoload$ is the sum of active power no-load losses of all elements downward the feeder.
- $LossQdownnoload$ is the sum of reactive power no-load losses of all elements downward the feeder.
- $du_feed$ is obtained using the magnitude of the voltage where the feeder is defined $u_{feeder}$ and the local positive sequence voltage magnitude:

$$du_feed = (u_{feeder} - u) \cdot 100$$

### 2.1.5 Max. current and loading for busbars

Table 2.5: Max. current and loading

| Name | Unit | Description |
|---|---|---|
| $Imax$ | kA | Max. Current |
| $loading$ | % | Loading |

The maximum current results for busbars are only available for busbars with a bay configuration. The topological order is defined by the position of the bays (see corresponding bay element (*ElmBay*) parameter *position*). A current is calculated for each section of the busbar. $Imax$ is then the maximum current of all sections.

If the busbar has a busbar type (*TypBar*) with a given rated current ($Ir$, see the Load Flow page) assigned, the loading for the busbar is calculated as follows:

$$loading = Imax/Ir \cdot 100$$

## 2.2 Result variables for elements

The result variables available for single- and multiple-port elements after a balanced Load Flow calculation are presented in the following sub chapters.

### 2.2.1 Voltage-related variables for elements

Similar to the voltage-related variables for terminals, these variables are calculated from the same positive-sequence, phase-to-ground complex voltage $\underline{u}$ resulting from the balanced load flow.

For the element based variables, the relationship between the absolute and per unit value voltage is $\underline{U} = \underline{u} \cdot U_{base}$ where the base voltage is $U_{base} = U_{nom_el}/\sqrt{3}$ where $U_{nom_el}$ is the nominal

line-to-line voltage of the element. Due to the change in base, the per unit values are multiplied with the factor $uknom/U_{nom_el}$.

Table 2.6: Voltage-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $u1$ | p.u. | Positive-Sequence-Voltage, Magnitude |
| $u1r$ | p.u. | Positive-Sequence Voltage, Real-Part |
| $u1i$ | p.u. | Positive-Sequence Voltage, Imaginary-Part |
| $phiu1$ | deg | Positive-Sequence-Voltage, Angle |
| $u$ | p.u. | Voltage, Magnitude |
| $ur$ | p.u. | Voltage, Real Part |
| $ui$ | p.u. | Voltage, Imaginary Part |
| $U1$ | kV | Line-Ground Positive-Sequence-Voltage, Magnitude |
| $U1l$ | kV | Line-Line Positive-Sequence-Voltage, Magnitude |

The result variables from Table 2.6 are calculated as follows:

- $u1$ is obtained from the complex voltage $\underline{u}$ as:

$$u1 = |\underline{u}| \cdot \frac{uknom}{U_{nom_el}}$$

  where $uknom$ is the nominal voltage of the connected terminal and $U_{nom_el}$ is the nominal voltage of the element.

- $u1r$ is obtained from the complex voltage $\underline{u}$ as:

$$u1r = \Re(\underline{u}) \cdot \frac{uknom}{U_{nom_el}}$$

- $u1i$ is obtained from the complex voltage $\underline{u}$ as:

$$u1i = \Im(\underline{u}) \cdot \frac{uknom}{U_{nom_el}}$$

- $phiu1$ is obtained from the complex voltage $\underline{u}$ as:

$$phiu = \arctan\left(\frac{\Im(\underline{u})}{\Re(\underline{u})}\right) \cdot \frac{180}{\pi}$$

- $phiurel$ is obtained using the *Initial Angle of Bus Voltage* $phiini$ basic data parameter as:

$$phiurel = phiu - phiini \cdot \frac{180}{\pi}$$

- $u$ is equal to the positive sequence voltage magnitude:

$$u = u1$$

- $ur$ is obtained as:

$$ur = u1r$$

- $ui$ is obtained as:

$$ui = u1i$$

- $U1$ is obtained as:

$$U1 = u \cdot U_{nom_el}/\sqrt{3}$$

- $U1l$ is obtained as:

$$U1l = u \cdot U_{nom_el}$$

### 2.2.2 Current-related variables for elements

The current-related variables for the elements are all based on the positive-sequence complex current resulting from the balanced load flow. The relationship between the absolute and per unit value current is $\underline{I} = \underline{i} \cdot I_{nom_el}$ where $I_{nom_el} = \frac{\mathrm{MVA}_{el}}{\sqrt{3} \cdot U_{nom_el}}$ is the nominal current of the element.

Table 2.7: Current-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $I1$ | kA | Positive-Sequence Current, Magnitude |
| $phii1$ | deg | Positive-Sequence Current, Angle |
| $I$ | kA | Current, Magnitude |
| $phii$ | deg | Current, Angle |
| $i1$ | p.u. | Positive-Sequence Current, Magnitude |
| $i1r$ | p.u. | Positive-Sequence Current, Real Part |
| $i1i$ | p.u. | Positive-Sequence Current, Imaginary Part |
| $i$ | p.u. | Current, Magnitude |
| $ir$ | p.u. | Current, Real Part |
| $ii$ | p.u. | Current, Imaginary Part |
| $i1P$ | p.u. | Positive-Sequence Active Current |
| $i1Q$ | p.u. | Positive-Sequence Reactive Current |
| $I1P$ | kA | Positive-Sequence Active Current |
| $I1Q$ | kA | Positive-Sequence Reactive Current |
| $phiui$ | deg | Angle between Voltage and Current |
| $phiu1i1$ | deg | Angle between Voltage and Current in positive sequence system |
| $inet$ | p.u. | Current, Magnitude, referred to network |

The result variables from Table 2.7 are calculated as follows:

- $I1$ is obtained as the amplitude of the complex current $\underline{I}$:

$$I1 = |\underline{I}|$$

- $phii1$ is obtained from the complex voltage $\underline{i}_{net}$ as:

$$phii1 = \arctan\left(\frac{\Im(\underline{I})}{\Re(\underline{I})}\right) \cdot \frac{180}{\pi}$$

- $I$ is equivalent to the positive sequence current magnitude:

$$I = I1$$

- $phii$ is obtained as:

$$phii = phii1$$

- $i1$ is obtained as:

$$i1 = \frac{I}{I_{nom_el}}$$

  where $I_{nom_el}$ is the nominal current of the element.

- $i1r$ is obtained as:

$$i1r = \frac{\Re(\underline{I})}{I_{nom_el}}$$

- $i1i$ is obtained as:

$$i1i = \frac{\Im(\underline{I})}{I_{nom_el}}$$

- $i$ is obtained as:

$$i = i1$$

- $ir$ is obtained as:

$$ir = i1r$$

- $ii$ is obtained as:

$$ii = i1i$$

- $I1P$, $I1Q$, $i1P$ and $i1Q$ are obtained as:

$$\begin{aligned} I1P &= I1 \cdot \cos(\varphi) = I1 \cdot cosphi \\ I1Q &= I1 \cdot \sin(\varphi) \\ i1P &= i1 \cdot \cos(\varphi) = \frac{I1P}{I_{nom_el}} \\ i1Q &= i1 \cdot \sin(\varphi) = \frac{I1Q}{I_{nom_el}} \end{aligned}$$

  where $\varphi$ is the angle of the complex apparent power and can be obtained as: $\varphi = \arccos(cosphi)$. See definition of $cosphi$ (power factor) in Section 2.2.3.

- $phiui$ is obtained as:

$$phiui = phiu - phii$$

- $phiu1i1$ is obtained as:

$$phiu1i1 = phiu1 - phii1$$

- $inet$ is obtained as:

$$inet = \frac{I}{I_{nom_1MVA}}$$

  where $I_{nom_1MVA} = \frac{1}{\sqrt{3} \cdot uknom}$ is the nominal current for 1 MVA and $uknom$ is the nominal voltage of the connected terminal.

### 2.2.3 Power-related variables for elements

The apparent power is calculated as $\underline{S} = 3 \cdot \underline{U}_1 \cdot \underline{I}^*$ where $\underline{U}_1$ is the positive sequence voltage.

Table 2.8: Power-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $S$ | MVA | Apparent Power |
| $P$ | MW | Active Power |
| $Q$ | Mvar | Reactive Power |
| $cosphi$ | | Power Factor |
| $tanphi$ | | tan(phi) |
| $Psum$ | MW | Total Active Power |
| $Qsum$ | Mvar | Total Reactive Power |
| $Ssum$ | MVA | Total Apparent Power |
| $cosphisum$ | | Total Power Factor |
| $tanphisum$ | | Total tan(phi) |
| $Spu$ | MVA/p.u. | Apparent Power per p.u. Voltage |

The result variables from Table 2.8 are calculated as follows:

- $S$ is obtained as the magnitude of the apparent power:

$$S = |\underline{S}|$$

- $P$ is obtained as:

$$P = \Re(\underline{S})$$

- $Q$ is obtained as:

$$Q = \Im(\underline{S})$$

- $cosphi$ is obtained as:

$$cosphi = \cos\left(\arctan\left(\frac{Q}{P}\right)\right) = \frac{P}{S}$$

- $tanphi$ is obtained as:

$$tanphi = \frac{Q}{P}$$

- $Psum$ is equal to the magnitude of the active power:

$$Psum = P$$

- $Qsum$ is equal to the magnitude of the reactive power:

$$Qsum = Q$$

- $Ssum$ is equal to the magnitude of the apparent power:

$$Ssum = S$$

- $cosphisum$ is obtained as:

$$cosphisum = cosphi$$

- $tanphisum$ is obtained as:

$$tanphisum = tanphi$$

- $Spu$ is obtained as:

$$Spu = \sqrt{3} \cdot uknom \cdot I$$

### 2.2.4 Miscellaneous variables for elements

Table 2.9: Miscellaneous variables for elements

| Name | Unit | Description |
|---|---|---|
| $Tfct$ | s | Fault Clearing Time |
| $Brkload$ | % | Breaker Loading |
| $Imax$ | kA | Maximum Current |
| $Smax$ | MVA | Maximum Power |

The result variables from Table 2.9 are calculated as follows:

- $Tfct$ gives the fault clearing time of a fuse or a relay located in the local cubicle. If the fuse/relay model is not triggered with the Load Flow current, a default value of $9999{,}999s$ is used.
- $Brkload$ is obtained as:

$$Brkload = \frac{I}{BrkInom} \cdot 100$$

  where $BrkInom$ is the rated current of a switch (input parameter $Inom$ in *TypSwitch*).
- $Imax$ is the maximum current from low-voltage load flow analysis (coincidence of low-voltage loads is considered).
- $Smax$ is obtained as:

$$Smax = \sqrt{3} \cdot uknom \cdot Imax$$

## 2.3 Result variables for grouping elements

### 2.3.1 Total results

Table 2.10: Total variables (generation and consumption)

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $TotgenP$ | MW | Total Generation, Active Power | Total P gen. |
| $TotgenQ$ | Mvar | Total Generation, Reactive Power | Total Q gen. |
| $TotconsP$ | MVA | Total Consumption, Active Power | Total P cons. |
| $TotconsQ$ | | Total Consumption, Reactive Power | Total Q cons. |
| $TotP$ | MW | Total Demand | |
| $TotLoadP$ | MW | Total Load, Active Power | |
| $TotLoadQ$ | Mvar | Total Load, Reactive Power | |
| $TotLoadS$ | MVA | Total Load, Apparent Power | |
| $TotLoadcos$ | Mvar | Total Load, Power Factor | |
| $UnsupP$ | MW | Unsupplied Demand | |
| $SupP$ | MW | Supplied Demand | |

The "Total Generation, Active Power" ($TotgenP$), is the sum of all active power fed into the system; that is, for generators when $P > 0$, and for loads when $P < 0$. Correspondingly, the reactive power ($TotgenQ$) is the sum of all reactive power infeed to the system; e.g. for generators when $Q > 0$, and for loads when $Q < 0$.

The "Total Consumption, Active Power" ($TotconP$), is the sum of all active power consumption from the system; e.g. for loads when $P > 0$, and for generators when $P < 0$.
Correspondingly, the reactive power is the sum of all reactive power consumption from the system; e.g. for loads when $Q > 0$, and for generators when $Q < 0$.

Notes:

- The losses of compensation models (see *Compensation (Losses)* in Table 2.16) are not taken into account for the active power consumption.
- The reactive power of rectifier/inverter models (*ElmRecmono*, *ElmRec* or *ElmHvdclcc*) are taken into account for the reactive power consumption.
- The reactive power of PWM Converters (*ElmVscmono* or *ElmVsc*) are dependent on the sign of Q, and are therefore assigned to the generation (when $Q > 0$), or to the consumption (when $Q < 0$), accordingly.

The demand quantities ($TotP$,$UnsupP$ and $SupP$) take all load models and all motors into account. The total demand ($TotP$) and total load, active and reactive power is then:

$$TotP = LoadPnom + MotPnom$$
$$TotLoadP = LoadP + MotP$$
$$TotLoadQ = LoadQ + MotQ$$

where:

- $LoadPnom$ is the nominal power of all load models; see Table 2.15
- $MotPnom$ is the sum of the nominal active power of all motors

- $LoadP$, $LoadQ$ is the active and reactive power of all load models; see Table 2.15
- $MotP$, $MotQ$ is the active and reactive power of all motor models; see Table 2.14

The active power is split into unsupplied and supplied loads and motors.

The "Total Load, Apparent Power", ($TotLoadS$), is then $TotLoadS = \sqrt{TotLoadP^2 + TotLoadQ^2}$ and the power factor is $TotLoadcos = TotLoadP/TotLoadS$.

For zones (*ElmZone*) and areas (*ElmArea*) the following additional quantities are available:

Table 2.11: Reactive power capacity related variables

| Name | Unit | Description |
|---|---|---|
| $RUC_Lead$ | Mvar | Used Reactive Capacity Lead |
| $RUC_Lag$ | Mvar | Used Reactive Capacity Lag |
| $RSC_Lead$ | Mvar | Spare Reactive Capacity Lead |
| $RSC_Lag$ | Nvar | Spare Reactive Capacity Lag |
| $Q_shgain$ | Mvar | Reactive Power Shunt Gain |
| $ConnMvar_max$ | Mvar | Connected MVAR Capacity (max) |
| $ConnMvar_min$ | Mvar | Connected MVAR Capacity (min) |

The used reactive capacity lead and lag is:

$$RUC_Lead = \sum Q_{gen} - Q_{comp,L} \quad \text{when the sign of } Q_{gen} \text{ is the same as that of } P_{gen}$$
$$RUC_Lag = \sum Q_{gen} - Q_{comp,C} \quad \text{when the sign of } Q_{gen} \text{ is different to that of } P_{gen}$$

where

- $Q_{comp,L}$ is the reactive power consumption of shunts/filter and SVS (positive for reactors);
- $Q_{comp,C}$ is the reactive power infeed to the system of shunts/filters and SVSs (negative for capacitors);
- $Q_{gen}$ is the reactive power of the generator model;
- $P_{gen}$ is the active power of the generator model.

If $P_{gen}$ is zero ($< 0.0001$ MW) and the generator is feeding reactive power into the system, the reactive power is added to the $RUC_Lead$ quantity, otherwise to the $RUC_Lag$ quantity.

The connected MVAR capacity (min,max) (only for spinning machines, shunts/filters and SVSs) is:

$$ConnMvar_max = \sum Q_{max}$$
$$ConnMvar_min = \sum Q_{min}$$

Is should be noted that $Q_{max}$ is positive for all capacitive shunts/filters and $Q_{min}$ is negative for all reactive shunts/filters.

The following generator models are taken into account:

- Synchronous generators (*ElmSym*);

- Asynchronous generators (*ElmAsm*) configured as a DFIG;
- Static generators (*ElmGenstat*);
- PV systems (*ElmPvsys*);

The spare reactive capacity is then the difference between the min., max. reactive power of the generators, shunts/filters and SVSs, and the actual reactive power:

$$RSC_Lead = \sum Q_{gen} - Q_{min}$$
$$RSC_Lag = \sum Q_{max} - Q_{gen}$$

The reactive power shunt gain, ($Q_shgain$), is the sum of the actual reactive power for shunts/filters (*ElmShnt*).

### 2.3.2 Sum results for generators

The following elements are taken into account:

- Synchronous Generators (*ElmSym*, configured as a generator)
- Asynchronous Generators (*ElmAsm*, configured as a generator)
- Static generators (*ElmGenstat*)
- PV Systems (*ElmPvsys*)
- DC Generators (with one or two terminals) (*ElmDcm* and *ElmDcmbi*, configured as a generator)
- DFIG Generators (*ElmAsmsc*, configured as a generator)
- MV loads (*ElmLodmv*, only the generation part

Table 2.12: Sum-related variables for generators

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $GenP$ | MW | Generators, Active Power | Generators, P |
| $GenQ$ | Mvar | Generators, Reactive Power | Generators, Q |
| $GenS$ | MVA | Generators, Apparent Power | Generators, S |
| $Gencos$ | | Generators, Power Factor | |
| $GenPnom$ | MW | Generators, Nominal Active Power | |
| $GenQnom$ | Mvar | Generators, Nominal Reactive Power | |
| $GenSnom$ | MVA | Generators, Nominal Apparent Power | |
| $Gencosnom$ | | Generators, Nominal Power Factor | |
| $GenPdiff$ | MW | Generators, difference between nominal and actual active power | |
| $GenQdiff$ | Mvar | Generators, difference between nominal and actual reactive power | |
| $GenPdisp$ | MW | Generators, Dispatch Active Power | |
| $GenPmism$ | MW | Generators, difference between actual and dispatched active power | |
| $GenPspinres$ | MW | Generators, difference between maximum and actual active power | |
| $GenQspinres$ | Mvar | Generators, difference between maximum and actual reactive power | |
| $cst_disp$ | X[1]/h | Total Production Costs | |

[1]: X is replaced by the currency unit of the project (USD, EUR, GBP, ... ).

$GenP$ and $GenQ$ is the sum of the active and reactive power of all generators assigned to the grouping object, regardless of the sign; e.g. if the generator consumes power (negative active or reactive power).

The apparent power, $GenS$, is then $GenS = \sqrt{GenP^2 + GenQ^2}$ and the power factor is $Gencos = GenP/GenS$.

The nominal active and reactive power ($GenPnom$ and $GenQnom$) are calculated based on the rated apparent power and rated power factor of the generators. The difference between the nominal and actual active and reactive power ($GenPdiff$ and $GenQdiff$ is then $GenPdiff = GenPnom - GenP$ and $GenPdiff = GenQnom - GenQ$.

The dispatched active power $GenPdisp$ is the sum of the entered dispatched active power, including the wind generation scaling factor of the zone (*ElmZone*) and the generator scaling factor from the Load Flow command. The difference between the actual and dispatched active power is then $GenPmism = GenP - GenPdisp$.

The difference between the maximum and actual active, reactive power ($GenPspinres$ and $GenQspinres$) is only for generators with the option *Consider for region spinning reserve* enabled (see the Load Flow page ⇒ Advanced). The external grid (*ElmXnet*) is also taken into account.
Note: The option is enabled by default for synchronous machines and disabled for static generators, PV systems, asynchronous machines and external grids. DFIG machines and MV loads (*ElmAsmsc* and *ElmLodmv*) are not taken into account.

$$GenPspinres = \sum (P_{max} - P_{gen})$$
$$GenQspinres = \sum (Q_{max} - Q_{gen})$$

where $P_{gen}$ is the active power, $Q_{gen}$ the reactive power and $P_{max}$, $Q_{max}$ are the maximum active and reactive power of the generator, respectively.

The total production cost ($cst_disp$) is calculated as follows and only considers synchronous generators (*ElmSym*), static generators (*ElmGenstat*, external grids (*ElmXnet*) and asynchronous generators (*ElmAsm*) configured as a DFIG.

### 2.3.3 Sum results for external networks

The following elements are taken into account:

- External Grids (*ElmXnet*)
- AC Voltage Sources (with one or two terminals) (*ElmVac* and *ElmVacbi*), but only if not used as an earthing model ($voltage\ setpoint < 1.^{-6}$ ).
- DC Voltage Sources (with one or two terminals) (*ElmDcu* and *ElmDcubi*), but only if not used as an earthing model ($|voltage\ setpoint| < 1.^{-6}$ ).
- Batteries (with one or two terminals) (*ElmBattery* and *ElmBatterybi*)

Table 2.13: Sum-related variables for external networks

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $NetP$ | MW | External Networks, Active Power | External Networks, P |
| $NetQ$ | Mvar | External Networks, Reactive Power | External Networks, Q |
| $NetS$ | MVA | External Networks, Apparent Power | External Networks, S |
| $Netcos$ | | External Networks, Power Factor | |

$NetP$ and $NetQ$ is the sum of the active and reactive power of all external network models assigned to the grouping object, regardless of the sign (consumption or infeed of power).

The apparent power $NetS$ is then $NetS = \sqrt{NetP^2 + NetQ^2}$ and the power factor is $Netcos = NetP/NetS$.

### 2.3.4 Sum results for motors

The following elements are taken into account:

- Synchronous Motors (*ElmSym*, configured as a motor)
- Asynchronous Motors (*ElmAsm*, configured as a motor)
- DC Motors (with one or two terminals) (*ElmDcm* and *ElmDcmbi*, configured as a motor)
- DFIG Motors (*ElmAsmsc*, configured as a motor)

Table 2.14: Sum-related variables for motors

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $MotP$ | MW | Motor Loads, Active Power | Motors, P |
| $MotQ$ | Mvar | Motor Loads, Reactive Power | Motors, Q |
| $MotS$ | MVA | Motor Loads, Apparent Power | Motors, S |
| $Motcos$ | | Motor Loads, Power Factor | |

The variables $MotP$ and $MotQ$ are the sum of the active and reactive power of all motors assigned to the grouping object, respectively, regardless of the sign; e.g. if the motor infeeds power (negative active or reactive power).

The apparent power, $MotS$, is then $MotS = \sqrt{MotP^2 + MotQ^2}$ and the power factor is $Motcos = MotP/MotS$.

### 2.3.5 Sum results for loads

The following elements are taken into account:

- General Loads (*ElmLod*)
- MV Loads (*ElmLodmv*, load part only)
- LV Loads (*ElmLodlv*)
- DC Loads (with one or two terminals) (*ElmLoddc* and *ElmLoddcbi*)

- Trains (AC or DC) (*ElmTrain*)
- AC Current Sources (with one or two terminals) ((*ElmIac* and *ElmIacbi*)
- DC Current Sources (with one or two terminals) ((*ElmDci* and *ElmDcibi*)
- Impulse Sources (*ElmImpulse*)
- Lines with line loads (*ElmLne* with *ElmLodlvp*)

Table 2.15: Sum-related variables for loads

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $LoadP$ | MW | Loads, Active Power | Loads, P |
| $LoadQ$ | Mvar | Loads, Reactive Power | Loads, Q |
| $LoadS$ | MVA | Loads, Apparent Power | Loads, S |
| $Loadcos$ | | Loads, Power Factor | |
| $LoadPnom$ | MW | Loads, Nominal Active Power | Loads, P |
| $LoadQnom$ | Mvar | Loads, Nominal Reactive Power | Loads, Q |
| $LoadSnom$ | MVA | Loads, Nominal Apparent Power | Loads, S |
| $Loadcosnom$ | | Loads, Nominal Power Factor | |
| $LoadPdiff$ | MW | Loads, difference between nominal and actual active power | Loads, dP(Un-U) |
| $LoadQdiff$ | Mvar | Loads, difference between nominal and actual reactive power | Loads, dQ(Un-U) |
| $LoadPdem$ | MW | Loads, Active Power Demand | |
| $LoadPmism$ | MW | Loads, difference between actual active power and demand | |

$LoadP$ and $LoadQ$ are the sum of the active and reactive power of all loads assigned to the grouping object, respectively, regardless of the sign; e.g. if the load infeeds power (negative active or reactive power).

The apparent power, $LoadS$, is then $LoadS = \sqrt{LoadP^2 + LoadQ^2}$ and the power factor is $Loadcos = LoadP/LoadS$.

The nominal active and reactive power, $LoadPnom$ and $LoadQnom$, respectively, are calculated based on the dispatched active and reactive power of the loads, including the entered scaling factor (*scale0*), the load scaling factor of the corresponding zone (*ElmZone*) and the load scaling factor of the Load Flow command.

The difference between the nominal and actual active, reactive power, $LoadPdiff$ and $LoadQdiff$, is then $LoadPdiff = LoadPnom - LoadP$ and $LoadPdiff = LoadQnom - LoadQ$.
It should be noted that the difference can be caused by, for example, the load flow being calculated using the *Distributed slack by load* option, or when the loads are voltage dependent or scaled via feeder load scaling.

The active power demand for loads ($LoadPdem$) is equal to the nominal active power of the loads. The differences between the actual active power and demand, $LoadPmism$, is then $LoadPmism = LoadP - LoadPdem$.

### 2.3.6 Sum results for compensation equipment

The following elements are taken into account:

- Static Var System (*ElmSvs*)
- Shunt/Filters, AC and DC (*ElmShnt*)
- Line compensation (*ElmLnecomp*)
- Harmonic Filters (*ElmFilter*)
- AC Voltage Sources (with one terminal) (*ElmVac*), only if used as an earthing model ($voltage\ setpoint < 1.^{-6}$ ).
- DC Voltage Sources (with one or two terminals) (*ElmDcu* and *ElmDcubi*), only if used as an earthing model ($|voltage\ setpoint| < 1.^{-6}$ ).

Table 2.16: Sum-related variables for compensation

| Name | Unit | Description |
|---|---|---|
| $CompP$ | MW | Compensation (Losses) |
| $CompQ$ | Mvar | Compensation, C |
| $ComprQ$ | Mvar | Compensation, L |

The compensation losses, $CompP$, are the total losses of all compensation elements assigned to the grouping object (i.e. the sum of the active power).

Compensation, C ($CompQ$) is the sum of the reactive power of all compensation elements assigned to the grouping object when $Q < 0$ (i.e. feeding reactive power into the system).

Compensation, L ($CompQr$) is the sum of the reactive power of all compensation elements assigned to the grouping object when $Q > 0$ (i.e. consuming reactive power from the system).

### 2.3.7 Interchange flow, Import and Export

Table 2.17: Interchange flow variables

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $ExportP$ | MW | Export, Active Power | Export, P |
| $ExportQ$ | Mvar | Export, Reactive Power | Export, Q |
| $ImportP$ | MW | Import, Active Power | Import, P |
| $ImportQ$ | Mvar | Import, Reactive Power | Import, Q |
| $InterP$ | MW | Interchange Flow, Active Power | Interchange Flow, P |
| $InterQ$ | Mvar | Interchange Flow, Reactive Power | Interchange Flow, Q |
| $ptot$ | MW | Total Area Exchange Power | |

For the export and import active and reactive power ($ExportP$, $ExportQ$, $ImportP$, $ImportQ$), all models are considered where the grouping is different; e.g. for lines, transformers between different grids, zones or areas. Elements with only one terminal connection are taken into account when the corresponding terminal is in a different grouping to that of the element. It should be noted that the export power and import power quantities are always positive.

The interchange flow of the active and reactive power, $InterP$ and $InterQ$, is positive when the power is exported and calculated as follows:

$$InterP = ExportP - ImportP$$
$$InterQ = ExportQ - ImportQ$$

The total area exchange power, $ptot$, is equal to the interchange flow active power, $InterP$, and is only available for grids (*ElmNet*).

### 2.3.8 Losses

The losses for elements with more than one connection are distributed depending on the option *Loss assignment* (see the Load Flow page) when the model is connected between two different grouping elements; e.g. inter grid/area/zone lines, transformers, etc).

The losses are assigned as follows:

- According to grouping: the losses are assigned to the grouping element;
- Uniformly distributed: the losses are split and assigned 50/50;
- To terminal i/j (HV,LV...): the losses are assigned to the corresponding grouping of the terminal.

#### 2.3.8.1 Total Losses

Table 2.18: Total losses

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $LossP$ | MW | Losses, Active Power | Losses, P |
| $LossQ$ | Mvar | Losses, Reactive Power | Losses, Q |
| $LossPld$ | MW | Losses, Active Power (load) | Losses, P (load) |
| $LossQld$ | Mvar | Losses, Reactive Power (load) | Losses, Q (load) |
| $LossPnld$ | MW | Losses, Active Power (no load) | Losses, P (no load) |
| $LossQnld$ | Mvar | Losses, Reactive Power (no load) | Losses, Q (no load) |

Losses are calculated for elements with more than one connection as the sum of the active power for all connections:

$$Ploss = \sum_{i=1}^{n} P_{bus(i)}$$

$$Qloss = \sum_{i=1}^{n} Q_{bus(i)}$$

where $n$ is the number of terminal connections.

The total losses are then the sum of all model losses for the corresponding assigned grouping element, depending on the *Loss assignment* parameter.

Note:

- For lines with line loads, the active and reactive power of the loads are not taken into account.
- Models with AC and DC connections (e.g. rectifier/inverter models (*ElmRecmono*, *ElmRec* or *ElmHvdclcc*) or PWM converters (*ElmVscmono* or *ElmVsc*)) are not taken into account for the reactive power losses ($LossQ$).

- Voltage and current source models and DC machines with two connections are not taken into account.
- *Compensation (Losses)* ($CompP$) (see Section 2.3.6) are also taken into account for the active power losses ($LossQ$).

The losses, active and reactive power (no load) are the losses for branch elements with a shunt; e.g. the iron losses, no load current (magnetising reactance) for transformers or for lines the with a given insulation factor or conductance.

The losses, active and reactive power (load) ($LossPld$ and $LossQld$) are then calculated as follows:

$$LossPld = LossP - LossPnld$$
$$LossQld = LossQ - LossQnld$$

Note:

- *Compensation (Losses)* ($CompP$) (see Section 2.3.6) are always taken into account for the losses, active power (load) ($LossPld$). For compensation models the no load losses equal zero.

#### 2.3.8.2 Line Losses

Line losses are calculated for all AC or DC line models (*ElmLne*).

Table 2.19: Line losses

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $LossPlne$ | MW | Line Losses, Active Power | Line Losses, P |
| $LossQlne$ | Mvar | Line Losses, Reactive Power | Line Losses, Q |
| $LossPldlne$ | MW | Line Losses, Active Power (load) | Line Losses, P (load) |
| $LossQldlne$ | Mvar | Line Losses, Reactive Power (load) | Line Losses, Q (load) |
| $LossPnldlne$ | MW | Line Losses, Active Power (no load) | Line Losses, P (no load) |
| $LossQnldlne$ | Mvar | Line Losses, Reactive Power (no load) | Line Losses, Q (no load) |

Losses for a line are calculated as follows:

$$Ploss = \sum_{i=1}^{2} P_{bus(i)} - P_{load} - P_{comp,i}$$
$$Qloss = \sum_{i=1}^{2} Q_{bus(i)} - Q_{load} - Q_{comp,i}$$

The total line losses are then the sum of all line losses for the corresponding assigned grouping element, depending on the *Loss assignment* parameter.

Note:

- The active and reactive power of the defined line loads ($P_{load}$, $Q_{load}$) are not taken into account for the line loss calculation.

- The active and reactive power of line compensation ($P_{comp,i}$, $Q_{comp,i}$) are not taken into account for the line loss calculation.

The line losses, active and reactive power (no load) ($LossPnldlne$ and $LossQnldlne$) are calculated as follows:

$$LossPnldlne = \sum Gload/1000$$
$$LossQnldlne = \sum -Cload$$

where:

- $Gload$ is the *Active no load losses, total* of a line in kW
- $Cload$ is the *Capacitive Losses, total* of a line in Mvar

For details please see Technical Reference of the Overhead Line Models, Section 'Losses'.

The losses, active and reactive power (load) ($LossPldlne$ and $LossQldlne$) are then calculated as follows:

$$LossPldlne = LossPlne - LossPnldlne$$
$$LossQldlne = LossQlne - LossQnldlne$$

#### 2.3.8.3 Transformer Losses

Transformer losses are calculated for all transformer models (*ElmTr2*, *ElmTr3*, *ElmTr4*, *ElmTrb*, *ElmVoltreg*), and additionally for MV loads (*ElmLodmv*) when a transformer is specified in the corresponding distribution transformer type (*TypDistrf*).

Table 2.20: Transformer losses

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $LossPtrf$ | MW | Transformer Losses, Active Power | Transformer Losses, P |
| $LossQtrf$ | Mvar | Transformer Losses, Reactive Power | Transformer Losses, Q |
| $LossPldtrf$ | MW | Transformer Losses, Active Power (load) | Transformer Losses, P (load) |
| $LossQldtrf$ | Mvar | Transformer Losses, Reactive Power (load) | Transformer Losses, Q (load) |
| $LossPnldtrf$ | MW | Transformer Losses, Active Power (no load) | Transformer Losses, P (no load) |
| $LossQnldtrf$ | Mvar | Transformer Losses, Reactive Power (no load) | Transformer Losses, Q (no load) |

Losses are calculated for all transformers as the sum of the active power for all connections:

$$Ploss = \sum_{i=1}^{n} P_{bus(i)}$$
$$Qloss = \sum_{i=1}^{n} Q_{bus(i)}$$

where $n$ is the number of terminal connections.

The total transformer losses are then the sum of all transformer losses for the corresponding assigned grouping element, depending on the *Loss assignment* parameter.

Note:

- In addition, the losses for the MV load with distribution transformers are also taken into account. For details please see Technical Reference of the MV Load, Section 'Calculation Quantities'.

The transformer losses, active and reactive power (no load) ($LossPnldtrf$ and $LossQnldtrf$) are calculated as follows:

$$LossPnldtrf = \sum Gmload/1000$$
$$LossQnldtrf = \sum Xmload/1000$$

where:

- $Gmload$ is the *Iron Losses* of a transformer or distribution transformer of an MW load in kW
- $Xmload$ is the *Magnetising Losses* of a transformer or distribution transformer of an MW load in kvar

For further details please see the Technical Reference of the Two-Winding Transformer (3-Phase), Chapter 'Losses'.

The losses, active and reactive power (load) ($LossPldtrf$ and $LossQldtrf$) are then calculated as follows:

$$LossPldtrf = LossPtrf - LossPnldtrf$$
$$LossQldtrf = LossQtrf - LossQnldtrf$$

### 2.3.9 Statistics

Table 2.21: Statistics

| Name | Description | Classes |
|---|---|---|
| $n_ele$ | Number of Elements | |
| $n_area$[1] | Number of Connected Grids | |
| $n_volt$ | Number of Voltage Levels | |
| $n_sta$[2] | No. of Substations | *ElmSubstat* and *ElmTrfstat* |
| $n_bar$ | No. of Busbars | *ElmTerm*, Usage: Busbar |
| $n_term$ | No. of Terminals | *ElmTerm*, Usage: Junction Node or Internal Node |
| $n_lne$ | No. of Lines | *ElmLne* |
| $n_2wt$[2] | No. of 2-w Trfs. | *ElmTr2* and *ElmTrb* |
| $n_3wt$[2] | No. of 3-w Trfs. | *ElmTr3* |
| $n_4wt$[2] | No. of 4-w Trfs. | *ElmTr4* |
| $n_sym$ | No. of syn. Machines | *ElmSym* |
| $n_asm$ | No. of asyn. Machines | *ElmAsm* and *ElmAsmsc* |
| $n_stg$ | No. of Static Generators | *ElmGenstat* |
| $n_pvs$ | No. of PV Systems | *ElmPvsys* |
| $n_lod$ | No. of Loads | *ElmLod*, *ElmLodmv*, *ElmLodlv*, *ElmLoddc*, *ElmLoddcbi*, *ElmIac*, *ElmIacbi*, *ElmDci*, *ElmDcibi*, *ElmTrain* |
| $n_svs$[2] | No. of SVS | *ElmSvs* |
| $n_shnt$[2] | No. of Shunts/Filters | *ElmShnt*, *ElmFilter* and *ElmVac*,*ElmDcu* used as an earthing element ($uset < 1^{-6}$) |
| $n_other$[2] | No. of Other Elements | |
| $n_cust$ | Number of Customers | *ElmLod*, *ElmLodmv*, *ElmLodlv*, *ElmLoddc*, *ElmLoddcbi*. In addition are line loads (*ElmLodlvp*) also taken into account |
| $n_cunsu$[2] | Number of unsupplied Customers | see $n_cust$ |
| $n_unsup$[2] | Number of unsupplied Loads | see $n_lod$ and in addition motors are also taken into account (*ElmSym*, *ElmAsm*, *ElmAsmsc*, *ElmDcm*, *ElmDcmbi*) |

Notes:

- [1]: For zones (*ElmZone*), a corresponding parameter is available: Number of Connected Zones, $n_zones$; and for areas, *ElmArea*: Number of Connected Areas, $n_areas$.
- [2]: Not available for zones (*ElmZone*) and areas (*ElmArea*).

In addition, the following statistics are available:

Table 2.22: Statistics

| Name | Description |
|---|---|
| $NrOvlBranch$ | Number of overloaded branch components |
| $NrGenQMax$ | Number of generators above their maximum reactive power limit |
| $NrGenQMin$ | Number of generators below their minimum reactive power limit |
| $NrTermMin$ | Number of terminals below their minimum voltage limit |
| $NrTermMax$ | Number of terminals above their maximum voltage limit |

where:

- $NrOvlBranch$ is the number of elements with more than one connection (pure neutral wire connections are not taken into account) where $loading > 100\%$
- $NrGenQMax$ is the number of synchronous generators (*ElmSym*), static generators (*ElmGenstat*) and PV systems (*ElmPvsys*) where $Q > Qmax + eps$
- $NrGenQMin$ is the number of synchronous generators (*ElmSym*), static generators (*ElmGenstat*) and PV systems (*ElmPvsys*) where $Q < Qmin - eps$
- $NrTermMin$ is the number of calculation nodes where $u < vmin$ ($vmin$, see the Load Flow page of the terminal (*ElmTerm*)). It should be noted that unsupplied nodes are not taken into account.
- $NrTermMax$ is the number of calculation nodes where $u > vmax$ ($vmax$, see the Load Flow page of the terminal (*ElmTerm*))

The variable $eps$ is the maximum acceptable allowed load flow error (see the Load Flow command), multiplied by 5.

The following are only available for grids (*ElmNet*):

Table 2.23: Overall statistics for summary grid

| Name | Description |
|---|---|
| $n_iso$ | No. of Isolated Areas |
| $n_isooff$ | No. of Unsupplied Isolated Areas |

An unsupplied isolated area is, for example, a terminal without a reference machine.

# 3 Unbalanced Load Flow calculation

The balanced load flow calculations uses multi-phase network representation which is valid for unbalanced networks. The resulting quantities are phase quantities.

For DC terminals and elements, the variables displaying imaginary values are zero.

## 3.1 Result variables for terminals

As described in Section 1.2.2, the result variables from the *Bus Results* set from the port elements are equivalent to the result variables from the *Currents, Voltages and Currents* set for the terminals.

The result variables available for terminals after an unbalanced Load Flow calculation are presented in the following sub chapters.

### 3.1.1 Phase voltage related variables for terminals

The voltage-related variables for terminals are all based on the phase-to-ground complex voltages $\underline{u}_A$, $\underline{u}_B$ and $\underline{u}_C$ resulting from the unbalanced load flow. The relationship between the absolute and per unit value voltage is $\underline{U}_A = \underline{u}_A \cdot U_{base}$ where the base voltage is $U_{base} = uknom/\sqrt{3}$ where $uknom$ is the nominal line-to-line voltage of the terminal.

Table 3.1: Phase voltage related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $ur$:$A$ | p.u. | Line-Ground Voltage, Real Part |
| $ur$:$B$ | p.u. | Line-Ground Voltage, Real Part |
| $ur$:$C$ | p.u. | Line-Ground Voltage, Real Part |
| $ui$:$A$ | p.u. | Line-Ground Voltage, Real Part |
| $ui$:$B$ | p.u. | Line-Ground Voltage, Real Part |
| $ui$:$C$ | p.u. | Line-Ground Voltage, Real Part |
| $u$:$A$ | p.u. | Line-Ground Voltage, Magnitude |
| $u$:$B$ | p.u. | Line-Ground Voltage, Magnitude |
| $u$:$C$ | p.u. | Line-Ground Voltage, Magnitude |
| $upc$:$A$ | % | Line-Ground Voltage, Magnitude |
| $upc$:$B$ | % | Line-Ground Voltage, Magnitude |
| $upc$:$C$ | % | Line-Ground Voltage, Magnitude |
| $U$:$A$ | kV | Line-Ground Voltage, Magnitude |
| $U$:$B$ | kV | Line-Ground Voltage, Magnitude |
| $U$:$C$ | kV | Line-Ground Voltage, Magnitude |
| $phiu$:$A$ | deg | Line-Ground Voltage, Angle |
| $phiu$:$B$ | deg | Line-Ground Voltage, Angle |
| $phiu$:$C$ | deg | Line-Ground Voltage, Angle |

The result variables from Table 3.1 are calculated as follows:

- $ur$:$A$, $ur$:$B$, $ur$:$C$ are the real part quantities of the resulting complex line-to-ground volt-

ages:

$$ur{:}A = \Re(\underline{u}_A)$$
$$ur{:}B = \Re(\underline{u}_B)$$
$$ur{:}C = \Re(\underline{u}_C)$$

- $ui{:}A$, $ui{:}B$, $ui{:}C$ are the imaginary part quantities of the resulting complex line-to-ground voltages:

$$ui{:}A = \Im(\underline{u}_A)$$
$$ui{:}B = \Im(\underline{u}_B)$$
$$ui{:}C = \Im(\underline{u}_C)$$

- $u{:}A$, $u{:}B$, $u{:}C$ are the magnitudes of the resulting line-ground voltages:

$$u{:}A = |\underline{u}_A|$$
$$u{:}B = |\underline{u}_B|$$
$$u{:}C = |\underline{u}_C|$$

- $upc{:}A$, $upc{:}B$, $upc{:}C$ are obtained as:

$$upc{:}A = u{:}A \cdot 100$$
$$upc{:}B = u{:}B \cdot 100$$
$$upc{:}C = u{:}C \cdot 100$$

- $U{:}A$, $U{:}B$, $U{:}C$ are obtained as:
  for AC terminals ($120°$):

$$U{:}A = u{:}A \cdot uknom/\sqrt{3}$$
$$U{:}B = u{:}B \cdot uknom/\sqrt{3}$$
$$U{:}C = u{:}C \cdot uknom/\sqrt{3}$$

  for AC/BI terminals ($180°$):

$$U{:}A = u{:}A \cdot uknom/2$$
$$U{:}B = u{:}B \cdot uknom/2$$
$$U{:}C = u{:}C \cdot uknom/2$$

  where $uknom$ is the nominal voltage of the terminal.

- $phiu{:}A$, $phiu{:}B$, $phiu{:}C$ are obtained as:

$$phiu{:}A = \arctan\left(\frac{ui{:}A}{ur{:}A}\right) \cdot \frac{180}{\pi}$$
$$phiu{:}B = \arctan\left(\frac{ui{:}B}{ur{:}B}\right) \cdot \frac{180}{\pi}$$
$$phiu{:}C = \arctan\left(\frac{ui{:}C}{ur{:}C}\right) \cdot \frac{180}{\pi}$$

### 3.1.2 Line-to-line voltage-related variables for terminals

The line-to-line voltages are calculated as the difference between the two phases:

for AC terminals ($120°$):

$$\underline{u}_{lA} = (\underline{u}_A - \underline{u}_B) / \sqrt{3}$$
$$\underline{u}_{lB} = (\underline{u}_B - \underline{u}_C) / \sqrt{3}$$
$$\underline{u}_{lC} = (\underline{u}_C - \underline{u}_A) / \sqrt{3}$$

for AC/BI terminals ($180°$):

$$\underline{u}_{lA} = (\underline{u}_A - \underline{u}_B) / 2$$
$$\underline{u}_{lB} = (\underline{u}_B - \underline{u}_C) / 2$$
$$\underline{u}_{lC} = (\underline{u}_C - \underline{u}_A) / 2$$

For a system containing two phases only the corresponding voltage phase difference is available ($\underline{u}_{lA}$ or $\underline{u}_{lB}$ or $\underline{u}_{lC}$).

For single phase systems, the line-to-line voltages are not available (cannot be calculated).

Table 3.2: Line-to-line voltage-related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $ul$:$A$ | p.u. | Line to Line Voltage, Magnitude |
| $ul$:$B$ | p.u. | Line to Line Voltage, Magnitude |
| $ul$:$C$ | p.u. | Line to Line Voltage, Magnitude |
| $ulpc$:$A$ | % | Line to Line Voltage, Magnitude |
| $ulpc$:$B$ | % | Line to Line Voltage, Magnitude |
| $ulpc$:$C$ | % | Line to Line Voltage, Magnitude |
| $Ul$:$A$ | kV | Line to Line Voltage, Magnitude |
| $Ul$:$B$ | kV | Line to Line Voltage, Magnitude |
| $Ul$:$C$ | kV | Line to Line Voltage, Magnitude |
| $phiul$:$A$ | deg | Line to Line Voltage, Angle |
| $phiul$:$B$ | deg | Line to Line Voltage, Angle |
| $phiul$:$C$ | deg | Line to Line Voltage, Angle |

The result variables from Table 3.2 are calculated as follows:

- $ul$:$A$, $ul$:$B$, $ul$:$C$ are the magnitudes of the line-to-line voltages:

$$ul{:}A = |\underline{u}_{lA}|$$
$$ul{:}B = |\underline{u}_{lB}|$$
$$ul{:}C = |\underline{u}_{lC}|$$

- $ulpc$:$A$, $ulpc$:$B$, $ulpc$:$C$ are obtained as:

$$ulpc{:}A = ul{:}A \cdot 100$$
$$ulpc{:}B = ul{:}B \cdot 100$$
$$ulpc{:}C = ul{:}C \cdot 100$$

- $Ul$:$A$, $Ul$:$B$, $Ul$:$C$ are obtained as:

$$Ul{:}A = ul{:}A \cdot uknom$$
$$Ul{:}B = ul{:}B \cdot uknom$$
$$Ul{:}C = ul{:}C \cdot uknom$$

  where $uknom$ is the nominal voltage of the terminal.

- $phiul$:$A$, $phiul$:$B$, $phiul$:$C$ are obtained as:

$$phiul{:}A = \arctan\left(\frac{\Im(\underline{u}_{lA})}{\Re(\underline{u}_{lA})}\right) \cdot \frac{180}{\pi}$$
$$phiul{:}B = \arctan\left(\frac{\Im(\underline{u}_{lB})}{\Re(\underline{u}_{lB})}\right) \cdot \frac{180}{\pi}$$
$$phiul{:}C = \arctan\left(\frac{\Im(\underline{u}_{lC})}{\Re(\underline{u}_{lC})}\right) \cdot \frac{180}{\pi}$$

### 3.1.3 Line-to-neutral voltage-related variables for terminals

The line-to-neutral voltages are calculated as the difference between the phase and neutral complex voltages:

$$\underline{u}_{lnA} = \underline{u}_A - \underline{u}_n$$
$$\underline{u}_{lnB} = \underline{u}_B - \underline{u}_n$$
$$\underline{u}_{lnC} = \underline{u}_C - \underline{u}_n$$

Table 3.3: Line-to-neutral voltage-related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $uln$:$A$ | p.u. | Line-Neutral Voltage, Magnitude |
| $uln$:$B$ | p.u. | Line-Neutral Voltage, Magnitude |
| $uln$:$C$ | p.u. | Line-Neutral Voltage, Magnitude |
| $Uln$:$A$ | kV | Line-Neutral Voltage, Magnitude |
| $Uln$:$B$ | kV | Line-Neutral Voltage, Magnitude |
| $Uln$:$C$ | kV | Line-Neutral Voltage, Magnitude |
| $phiuln$:$A$ | deg | Line-Neutral Voltage, Angle |
| $phiuln$:$B$ | deg | Line-Neutral Voltage, Angle |
| $phiuln$:$C$ | deg | Line-Neutral Voltage, Angle |
| $upht$:$A$ | p.u. | Phase Technology dependent Voltage, Magnitude |
| $upht$:$B$ | p.u. | Phase Technology dependent Voltage, Magnitude |
| $upht$:$C$ | p.u. | Phase Technology dependent Voltage, Magnitude |
| $Upht$:$A$ | kV | Phase Technology dependent Voltage, Magnitude |
| $Upht$:$B$ | kV | Phase Technology dependent Voltage, Magnitude |
| $Upht$:$C$ | kV | Phase Technology dependent Voltage, Magnitude |

If no neutral connection exists the values of the variables from Table 3.3 are set to zero. If neutral connection exists, the result variables are calculated as follows:

- $uln$:$A$, $uln$:$B$, $uln$:$C$ are the magnitudes of the line-to-neutral voltages:

$$uln{:}A = |\underline{u}_{lnA}|$$
$$uln{:}B = |\underline{u}_{lnB}|$$
$$uln{:}C = |\underline{u}_{lnC}|$$

- $Uln$:$A$, $Uln$:$B$, $Uln$:$C$ are obtained as:
  for AC terminals with neutral ($120°$):

$$Uln{:}A = uln{:}A \cdot uknom/\sqrt{3}$$
$$Uln{:}B = uln{:}B \cdot uknom/\sqrt{3}$$
$$Uln{:}C = uln{:}C \cdot uknom/\sqrt{3}$$

for AC/BI terminals with neutral ($180°$):

$$Uln{:}A = uln{:}A \cdot uknom/2$$
$$Uln{:}B = uln{:}B \cdot uknom/2$$
$$Uln{:}C = uln{:}C \cdot uknom/2$$

where $uknom$ is the nominal voltage of the terminal.

- $phiuln{:}A$, $phiuln{:}B$, $phiuln{:}C$ are obtained as:

$$phiuln{:}A = \arctan\left(\frac{\Im(\underline{u}_{lnA})}{\Re(\underline{u}_{lnA})}\right) \cdot \frac{180}{\pi}$$
$$phiuln{:}B = \arctan\left(\frac{\Im(\underline{u}_{lnB})}{\Re(\underline{u}_{lnB})}\right) \cdot \frac{180}{\pi}$$
$$phiuln{:}C = \arctan\left(\frac{\Im(\underline{u}_{lnC})}{\Re(\underline{u}_{lnC})}\right) \cdot \frac{180}{\pi}$$

- $upht{:}A$, $upht{:}B$, $upht{:}C$ are obtained depending if there is neutral connection or not. If there is neutral connection the variables are obtained as:

$$upht{:}A = uln{:}A$$
$$upht{:}B = uln{:}B$$
$$upht{:}C = uln{:}C$$

If there is no neutral connection as:

$$upht{:}A = ul{:}A$$
$$upht{:}B = ul{:}B$$
$$upht{:}C = ul{:}C$$

- $Upht{:}A$, $Upht{:}B$, $Upht{:}C$ are obtained depending if there is neutral connection or not. If there is neutral connection the variables are obtained as:

$$Upht{:}A = Uln{:}A$$
$$Upht{:}B = Uln{:}B$$
$$Upht{:}C = Uln{:}C$$

If there is no neutral connection as:

$$Upht{:}A = Ul{:}A$$
$$Upht{:}B = Ul{:}B$$
$$Upht{:}C = Ul{:}C$$

### 3.1.4 0,1,2 sequence and neutral voltage-related variables for terminals

In addition to the phase, line-to-line and line-to-neutral voltage quantities, quantities in the positive, negative and zero sequence are available as well. $\underline{u}_1$, $\underline{u}_2$ and $\underline{u}_0$ are obtained when the phase values are transformed to symmetrical components.

For 3-phase terminals:

$$\underline{u}0 = \tfrac{1}{3}\left(\underline{u}_A + \underline{u}_B + \underline{u}_C\right)$$
$$\underline{u}1 = \tfrac{1}{3}\left(\underline{u}_A + a \cdot \underline{u}_B + a^2 \cdot \underline{u}_C\right)$$
$$\underline{u}2 = \tfrac{1}{3}\left(\underline{u}_A + a^2 \cdot \underline{u}_B + a \cdot \underline{u}_C\right)$$

where $a = \angle 120°$

For BI-phase terminals ($180°$):

$$\underline{u}0 = \frac{1}{2}(\underline{u}_A + \underline{u}_B)$$
$$\underline{u}1 = \frac{1}{2}(\underline{u}_A - \underline{u}_B)$$
$$\underline{u}2 = 0$$

For 2-phase terminals ($120°$):

$$\underline{u}0 = \frac{1}{\sqrt{3}}(\underline{u}_A + \underline{u}_B)$$
$$\underline{u}1 = \frac{1}{\sqrt{3}}(\underline{u}_A - \underline{u}_B)$$
$$\underline{u}2 = 0$$

For 1-phase terminals:

$$\underline{u}0 = 0$$
$$\underline{u}1 = \underline{u}_A$$
$$\underline{u}2 = 0$$

Table 3.4: 0,1,2 sequence voltage-related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $un$ | p.u. | Neutral-Ground Voltage, Magnitude |
| $Un$ | kV | Neutral-Ground Voltage, Magnitude |
| $phiun$ | deg | Neutral-Ground Voltage, Angle |
| $um$ | p.u. | Average-Voltage, Magnitude |
| $Um$ | kV | Average-Voltage, Magnitude |
| $u0$ | p.u. | Zero-Sequence Voltage, Magnitude |
| $U0$ | kV | Zero-Sequence Voltage, Magnitude |
| $U0\times3$ | kV | 3*U0 |
| $phiu0$ | deg | Zero-Sequence Voltage, Angle |
| $u1$ | p.u. | Positive-Sequence Voltage, Magnitude |
| $u1pc$ | % | Positive-Sequence Voltage, Magnitude |
| $u1r$ | p.u. | Positive-Sequence Voltage, Real Part |
| $u1i$ | p.u. | Positive-Sequence Voltage, Imaginary Part |
| $U1$ | kV | Line-Ground Positive-Sequence Voltage, Magnitude |
| $phiu1$ | deg | Positive-Sequence Voltage, Angle |
| $phiurel$ | deg | Voltage, Relative Angle |
| $u2$ | p.u. | Negative-Sequence Voltage, Magnitude |
| $U2$ | kV | Line-Ground Negative-Sequence Voltage, Magnitude |
| $phiu2$ | deg | Negative-Sequence Voltage, Angle |
| $U1l$ | kV | Line to Line Positive-Sequence Voltage, Magnitude |
| $U2l$ | kV | Line to Line Negative-Sequence Voltage, Magnitude |
| $uphtmin$ | p.u. | Minimum of Phase Technology dependent Voltage, Magnitude |
| $uphtmax$ | p.u. | Maximum of Phase Technology dependent Voltage, Magnitude |
| $ubfac$ | % | Unbalance factor |

The variables from Table 3.4 are calculated as follows:

- $un$ is obtained from the neutral complex voltage as:

$$un = |\underline{u}_n|$$

- $Un$ is obtained as:
  for AC terminals ($120°$):

$$Un = un \cdot uknom/\sqrt{3}$$

  for AC/BI terminals ($180°$):

$$Un = un \cdot uknom/2$$

  where $uknom$ is the nominal voltage of the terminal.

- $phiun$ is obtained as:

$$phiun = \arctan\left(\frac{\Im(\underline{u}_n)}{\Re(\underline{u}_n)}\right) \cdot \frac{180}{\pi}$$

- $um$ is calculated by dividing the sum of phase voltage magnitudes of all phases by the number of phases.

- $Um$ is obtained as:
  for AC terminals:

$$Um = um \cdot uknom/\sqrt{3}$$

  for AC/BI terminals:

$$Um = um \cdot uknom/2$$

  where $uknom$ is the nominal voltage of the terminal.

- $u0$ is obtained from the zero sequence complex voltage as:

$$u0 = |\underline{u}_0|$$

- $U0$ is obtained as:
  for AC terminals ($120°$):

$$U0 = u0 \cdot uknom/\sqrt{3}$$

  for AC/BI terminals ($180°$):

$$U0 = u0 \cdot uknom/2$$

  where $uknom$ is the nominal voltage of the terminal.

- $U0{\times}3$ is calculated as $3 \cdot U0$ for three phase, as $2 \cdot U0$ for two phase and as $U0$ for single phase systems.

- $phiu0$ is obtained as:

$$phiu0 = \arctan\left(\frac{\Im(\underline{u}_0)}{\Re(\underline{u}_0)}\right) \cdot \frac{180}{\pi}$$

- $u1$ is obtained from the positive sequence complex voltage as:

$$u1 = |\underline{u}_1|$$

- $u1pc$ is obtained as:

$$u1pc = u1 \cdot 100$$

- $u1r$ is obtained from the positive sequence complex voltage as:

$$u1r = \Re(\underline{u}_1)$$

- $u1i$ is obtained from the positive sequence complex voltage as:

$$u1i = \Im(\underline{u}_1)$$

- $U1$ is obtained as:
  for AC terminals ($120°$):

$$U1 = u1 \cdot uknom/\sqrt{3}$$

  for AC/BI terminals ($180°$):

$$U1 = u1 \cdot uknom/2$$

  where $uknom$ is the nominal voltage of the terminal.

- $phiu1$ is obtained as:

$$phiu1 = \arctan\left(\frac{\Im(\underline{u}_1)}{\Re(\underline{u}_1)}\right) \cdot \frac{180}{\pi}$$

- $phiurel$ is obtained by using the *Initial Angle of Bus Voltage* $phiini$ basic data parameter as:

$$phiurel = phiu_1 - phiini \cdot \frac{180}{\pi}$$

- $u2$ is obtained from the negative sequence complex voltage as:

$$u2 = |\underline{u}_2|$$

- $U2$ is obtained as (only for 3-phase terminals):

$$U2 = u2 \cdot uknom/\sqrt{3}$$

  where $uknom$ is the nominal voltage of the terminal.

- $phiu2$ is obtained as:

$$phiu2 = \arctan\left(\frac{\Im(\underline{u}_2)}{\Re(\underline{u}_2)}\right) \cdot \frac{180}{\pi}$$

- $U1l$ is obtained as:
  for 2-phase and 3-phase terminals:

$$U1l = u1 \cdot uknom$$

  for 1-phase terminals:

$$U1l = U1$$

- $uphtmin$ is the minimum of $upht$:$A$, $upht$:$B$, $upht$:$C$.
- $uphtmax$ is the maximum of $upht$:$A$, $upht$:$B$, $upht$:$C$.
- $ubfac$ is calculated as:

$$ubfac = \frac{u2}{u1} \cdot 100$$

  If number of phases is less then three, it is set to zero.

### 3.1.5 Aggregated power variables for terminals

Table 3.5: Aggregated power variables for terminals

| Name | Unit | Description |
|---|---|---|
| $Pgen$ | MW | Generation, Active Power |
| $Qgen$ | Mvar | Generation, Reactive Power |
| $Pmot$ | MW | Motor Load, Active Power |
| $Qmot$ | Mvar | Motor Load, Reactive Power |
| $Pload$ | MW | General Load, Active Power |
| $Qload$ | Mvar | General Load, Reactive Power |
| $Pcomp$ | MW | Compensation (Losses) |
| $Qcomp$ | Mvar | Compensation |
| $Pnet$ | MW | External Networks, Active Power |
| $Qnet$ | Mvar | External Networks, Reactive Power |
| $Pflow$ | MW | Power Flow, Active Power |
| $Qflow$ | Mvar | Power Flow, Reactive Power |
| $Pout$ | MW | Outgoing Flow, Active Power |
| $Qout$ | Mvar | Outgoing Power, Reactive Power |
| $Sout$ | MVA | Outgoing Power, Apparent Power |
| $cosphiout$ | | Outgoing Power, Power Factor |
| $Pbalance$ | MW | Active Power Balance (=0) |
| $Qbalance$ | Mvar | Reactive Power Balance (=0) |

The result variables from Table 3.5 are equivalent as the variables presented from the balanced load flow calculation in Table 2.2. Please refer to Table 2.2 for more information.

### 3.1.6 Low-voltage analysis related variables for terminals

Table 3.6: Low-voltage analysis related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $umin$ | p.u. | Minimum Voltage |
| $Umin$ | kV | Minimum Voltage (Line to Neutral) |
| $dumax$ | % | Maximum Voltage Drop along Feeder |

The result variables from Table 3.6 are calculated from a low-voltage load flow analysis where coincidence of low-voltage loads is considered.

### 3.1.7 Feeder losses variables for terminals

Table 3.7: Load flow, unbalanced, feeder losses variables (terminals)

| Name | Unit | Description |
|---|---|---|
| $LossPdown$ | MW | Losses, downstream |
| $LossQdown$ | Mvar | Losses, downstream (Reactive Power) |
| $LossPdownload$ | MW | Load losses, downstream |
| $LossQdownload$ | Mvar | Load losses, downstream |
| $LossPdownnoload$ | MW | No load losses, downstream |
| $LossQdownnoload$ | Mvar | No load losses, downstream |

Please refer to Table 2.4 for more information on the variables from Table 3.7.

### 3.1.8 Max. current and loading for busbars

Table 3.8: Max. current and loading

| Name | Unit | Description |
|---|---|---|
| $Imax$ | kA | Max. Current |
| $ImaxPh : A$ | kA | Max. Current (per phase) A |
| $ImaxPh : B$ | kA | Max. Current (per phase) B |
| $ImaxPh : C$ | kA | Max. Current (per phase) C |
| $ImaxPh : N$ | kA | Max. Current (per phase) N |
| $loading$ | % | Max. Loading |
| $loadingPh : A$ | % | Loading A |
| $loadingPh : B$ | % | Loading B |
| $loadingPh : C$ | % | Loading C |
| $loadingPh : N$ | % | Loading N |

The maximum current results for busbars are only available for busbars with a bay configuration. The topological order is defined by the position of the bays (see corresponding bay element (*ElmBay*) parameter *position*).
A current is calculated for each section and phase of the busbar. $ImaxPh$ (A/B/C/N) is then the maximum current of all sections per phase. The max. current, $Imax$, is the maximum of all max. currents per phase.

If the busbar has a busbar type (*TypBar*) with a given rated current ($Ir$, see the Load Flow page) assigned, the loading for the busbar is calculated as follows:

$$loading = Imax/Ir \cdot 100$$

## 3.2 Result variables for elements

The result variables available for for single- and multiple-port elements after an unbalanced Load Flow calculation are presented in the following sub chapters.

### 3.2.1 Voltage-related variables for elements

The voltage-related variables for terminals are all based on the phase-to-ground complex voltages $\underline{u}_A$, $\underline{u}_B$, $\underline{u}_C$ and $\underline{u}_N$ resulting from the unbalanced load flow.

For the element based variables, the relationship between the absolute and per unit value voltage is $\underline{U}_A = \underline{u}_A \cdot U_{base}$ where the base voltage is $U_{base} = U_{nom_el}/\sqrt{3}$ where $U_{nom_el}$ is the nominal line-to-line voltage of the element. Due to the change in base, the per unit values are multiplied with the factor $uknom/U_{nom_el}$.

Table 3.9: Voltage-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $ur$:$A$ | p.u. | Phase Voltage, Real Part |
| $ur$:$B$ | p.u. | Phase Voltage, Real Part |
| $ur$:$C$ | p.u. | Phase Voltage, Real Part |
| $ur$:$N$ | p.u. | Phase Voltage, Real Part |
| $ui$:$A$ | p.u. | Phase Voltage, Imaginary Part |
| $ui$:$B$ | p.u. | Phase Voltage, Imaginary Part |
| $ui$:$C$ | p.u. | Phase Voltage, Imaginary Part |
| $ui$:$N$ | p.u. | Phase Voltage, Imaginary Part |
| $u$:$A$ | p.u. | Phase Voltage, Magnitude |
| $u$:$B$ | p.u. | Phase Voltage, Magnitude |
| $u$:$C$ | p.u. | Phase Voltage, Magnitude |
| $u$:$N$ | p.u. | Phase Voltage, Magnitude |

The result variables from Table 3.9 are calculated as follows:

- $ur$:$A$, $ur$:$B$, $ur$:$C$, $ur$:$N$ are obtained from the complex phase voltages as:

$$ur{:}A = \Re(\underline{u}_A) \cdot \frac{uknom}{U_{nom_el}}$$
$$ur{:}B = \Re(\underline{u}_B) \cdot \frac{uknom}{U_{nom_el}}$$
$$ur{:}C = \Re(\underline{u}_C) \cdot \frac{uknom}{U_{nom_el}}$$
$$ur{:}N = \Re(\underline{u}_N) \cdot \frac{uknom}{U_{nom_el}}$$

  where $uknom$ is the nominal voltage of the connected terminal and $U_{nom_el}$ is the nominal voltage of the element.

- $ui$:$A$, $ui$:$B$, $ui$:$C$, $ui$:$N$ are obtained from the complex phase voltages as:

$$ui{:}A = \Im(\underline{u}_A) \cdot \frac{uknom}{U_{nom_el}}$$
$$ui{:}B = \Im(\underline{u}_B) \cdot \frac{uknom}{U_{nom_el}}$$
$$ui{:}C = \Im(\underline{u}_C) \cdot \frac{uknom}{U_{nom_el}}$$
$$ui{:}N = \Im(\underline{u}_N) \cdot \frac{uknom}{U_{nom_el}}$$

  where $uknom$ is the nominal voltage of the connected terminal and $U_{nom_el}$ is the nominal voltage of the element.

- $u$:$A$, $u$:$B$, $u$:$C$, $u$:$N$ are obtained from the complex phase voltages as:

$$u{:}A = |\underline{u}_A| \cdot \frac{uknom}{U_{nom_el}}$$
$$u{:}B = |\underline{u}_B| \cdot \frac{uknom}{U_{nom_el}}$$
$$u{:}C = |\underline{u}_C| \cdot \frac{uknom}{U_{nom_el}}$$
$$u{:}N = |\underline{u}_N| \cdot \frac{uknom}{U_{nom_el}}$$

### 3.2.2 Current-related variables for elements

The current-related variables for the elements are all based on the complex phase current resulting from the unbalanced load flow. The relationship between the absolute and per unit value current is $\underline{I}_A = \underline{i}_A \cdot I_{nom_el}$ where $I_{nom_el} = \frac{\mathrm{MVA}_{el}}{\sqrt{3} \cdot U_{nom_el}}$ is the nominal current of the element.

Table 3.10: Current-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $I$:$A$ | kA | Phase Current, Magnitude |
| $I$:$B$ | kA | Phase Current, Magnitude |
| $I$:$C$ | kA | Phase Current, Magnitude |
| $I$:$N$ | kA | Phase Current, Magnitude |
| $phii$:$A$ | deg | Phase Current, Angle |
| $phii$:$B$ | deg | Phase Current, Angle |
| $phii$:$C$ | deg | Phase Current, Angle |
| $phii$:$N$ | deg | Phase Current, Angle |
| $ir$:$A$ | p.u. | Phase Current, Real Part |
| $ir$:$B$ | p.u. | Phase Current, Real Part |
| $ir$:$C$ | p.u. | Phase Current, Real Part |
| $ir$:$N$ | p.u. | Phase Current, Real Part |
| $ii$:$A$ | p.u. | Phase Current, Imaginary Part |
| $ii$:$B$ | p.u. | Phase Current, Imaginary Part |
| $ii$:$C$ | p.u. | Phase Current, Imaginary Part |
| $ii$:$N$ | p.u. | Phase Current, Imaginary Part |
| $i$:$A$ | p.u. | Phase Current, Magnitude |
| $i$:$B$ | p.u. | Phase Current, Magnitude |
| $i$:$C$ | p.u. | Phase Current, Magnitude |
| $i$:$N$ | p.u. | Phase Current, Magnitude |
| $phiui$:$A$ | deg | Angle between Voltage and Current |
| $phiui$:$B$ | deg | Angle between Voltage and Current |
| $phiui$:$C$ | deg | Angle between Voltage and Current |
| $phiui$:$N$ | deg | Angle between Voltage and Current |
| $inet$:$A$ | p.u. | Phase Current, Magnitude, referred to network |
| $inet$:$B$ | p.u. | Phase Current, Magnitude, referred to network |
| $inet$:$C$ | p.u. | Phase Current, Magnitude, referred to network |
| $inet$:$N$ | p.u. | Phase Current, Magnitude, referred to network |

The result variables from Table 3.10 are calculated as follows:

- $I$:$A$, $I$:$B$, $I$:$C$, $I$:$N$ are obtained as amplitudes of the complex currents:

$$I{:}A = |\underline{I}_A|$$
$$I{:}B = |\underline{I}_B|$$
$$I{:}C = |\underline{I}_C|$$
$$I{:}N = |\underline{I}_N|$$

- $phii{:}A$, $phii{:}B$, $phii{:}C$, $phii{:}N$ are obtained as:

$$phii{:}A = \arctan\left(\frac{\Im(\underline{I}_A)}{\Re(\underline{I}_A)}\right) \cdot \frac{180}{\pi}$$
$$phii{:}B = \arctan\left(\frac{\Im(\underline{I}_B)}{\Re(\underline{I}_B)}\right) \cdot \frac{180}{\pi}$$
$$phii{:}C = \arctan\left(\frac{\Im(\underline{I}_C)}{\Re(\underline{I}_C)}\right) \cdot \frac{180}{\pi}$$
$$phii{:}N = \arctan\left(\frac{\Im(\underline{I}_N)}{\Re(\underline{I}_N)}\right) \cdot \frac{180}{\pi}$$

- $ir{:}A$, $ir{:}B$, $ir{:}C$, $ir{:}N$ are obtained as:

$$ir{:}A = \frac{\Re(\underline{I}_A)}{I_{nom_el}}$$
$$ir{:}B = \frac{\Re(\underline{I}_B)}{I_{nom_el}}$$
$$ir{:}C = \frac{\Re(\underline{I}_C)}{I_{nom_el}}$$
$$ir{:}N = \frac{\Re(\underline{I}_N)}{I_{nom_el}}$$

- $ii{:}A$, $ii{:}B$, $ii{:}C$, $ii{:}N$ are obtained as:

$$ii{:}A = \frac{\Im(\underline{I}_A)}{I_{nom_el}}$$
$$ii{:}B = \frac{\Im(\underline{I}_B)}{I_{nom_el}}$$
$$ii{:}C = \frac{\Im(\underline{I}_C)}{I_{nom_el}}$$
$$ii{:}N = \frac{\Im(\underline{I}_N)}{I_{nom_el}}$$

- $i{:}A$, $i{:}B$, $i{:}C$, $i{:}N$ are obtained as:

$$i{:}A = \frac{I{:}A}{I_{nom_el}}$$
$$i{:}B = \frac{I{:}B}{I_{nom_el}}$$
$$i{:}C = \frac{I{:}C}{I_{nom_el}}$$
$$i{:}N = \frac{I{:}N}{I_{nom_el}}$$

- $phiui{:}A$, $phiui{:}B$, $phiui{:}C$, $phiui{:}N$ are obtained as:

$$phiui{:}A = \arctan\left(\frac{\Im(\underline{u}_A)}{\Re(\underline{u}_A)}\right) \cdot \frac{180}{\pi} - phii{:}A$$
$$phiui{:}B = \arctan\left(\frac{\Im(\underline{u}_B)}{\Re(\underline{u}_B)}\right) \cdot \frac{180}{\pi} - phii{:}B$$
$$phiui{:}C = \arctan\left(\frac{\Im(\underline{u}_C)}{\Re(\underline{u}_C)}\right) \cdot \frac{180}{\pi} - phii{:}C$$
$$phiui{:}N = \arctan\left(\frac{\Im(\underline{u}_N)}{\Re(\underline{u}_N)}\right) \cdot \frac{180}{\pi} - phii{:}N$$

- $inet{:}A$, $inet{:}B$, $inet{:}C$, $inet{:}N$ are obtained from the amplitude of the phase currents as:

$$inet{:}A = \frac{I{:}A}{I_{nom_1\text{MVA}}}$$
$$inet{:}B = \frac{I{:}B}{I_{nom_1\text{MVA}}}$$
$$inet{:}C = \frac{I{:}C}{I_{nom_1\text{MVA}}}$$
$$inet{:}N = \frac{I{:}N}{I_{nom_1\text{MVA}}}$$

where $I_{nom_1\text{MVA}} = \frac{1}{\sqrt{3} \cdot uknom}$ is the nominal current for 1 MVA and $uknom$ is the nominal voltage of the connected terminal.

### 3.2.3 Miscellaneous variables per phase for elements

Table 3.11: Miscellaneous variables per phase for elements

| Name | Unit | Description |
|---|---|---|
| $TfctPh{:}A$ | s | Fault Clearing Time |
| $TfctPh{:}B$ | s | Fault Clearing Time |
| $TfctPh{:}C$ | s | Fault Clearing Time |
| $TfctPh{:}N$ | s | Fault Clearing Time |
| $BrkloadPh{:}A$ | % | Breaker Loading |
| $BrkloadPh{:}B$ | % | Breaker Loading |
| $BrkloadPh{:}C$ | % | Breaker Loading |
| $BrkloadPh{:}N$ | % | Breaker Loading |

The result variables from Table 3.11 are calculated as follows:

- $TfctPh{:}A$, $TfctPh{:}B$, $TfctPh{:}C$, $TfctPh{:}N$ give the fault clearing time of a fuse or a relay located in the local cubicle. If the fuse/relay model is not triggered with the Load Flow current, a default value of $9999{,}999s$ is used.
- $BrkloadPh{:}A$, $BrkloadPh{:}B$, $BrkloadPh{:}C$, $BrkloadPh{:}N$ are calculated as:

$$BrkloadPh{:}A = \frac{I{:}A}{BrkInom} \cdot 100$$
$$BrkloadPh{:}B = \frac{I{:}B}{BrkInom} \cdot 100$$
$$BrkloadPh{:}C = \frac{I{:}C}{BrkInom} \cdot 100$$
$$BrkloadPh{:}N = \frac{I{:}N}{BrkInom} \cdot 100$$

where $BrkInom$ is the rated current of a switch (input parameter $Inom$ in *TypSwitch*) located in the cubicle.

### 3.2.4 0,1,2 sequence voltage-related variables for elements

In addition to the phase, line-to-line and line-to-neutral voltage quantities, quantities in the positive, negative and zero sequence are available as well.

For 3-phase elements:

$$\underline{u}0 = \frac{1}{3}\left(\underline{u}_A + \underline{u}_B + \underline{u}_C\right)$$
$$\underline{u}1 = \frac{1}{3}\left(\underline{u}_A + a \cdot \underline{u}_B + a^2 \cdot \underline{u}_C\right)$$
$$\underline{u}2 = \frac{1}{3}\left(\underline{u}_A + a^2 \cdot \underline{u}_B + a \cdot \underline{u}_C\right)$$

where $a = \angle 120°$

For BI-phase elements ($180°$):

$$\underline{u}0 = \frac{1}{2}\left(\underline{u}_A + \underline{u}_B\right)$$
$$\underline{u}1 = \frac{1}{2}\left(\underline{u}_A - \underline{u}_B\right)$$
$$\underline{u}2 = 0$$

For 2-phase elements ($120°$):

$$\underline{u}0 = \frac{1}{\sqrt{3}}\left(\underline{u}_A + \underline{u}_B\right)$$
$$\underline{u}1 = \frac{1}{\sqrt{3}}\left(\underline{u}_A - \underline{u}_B\right)$$
$$\underline{u}2 = 0$$

For 1-phase elements:

$$\underline{u}0 = 0$$
$$\underline{u}1 = \underline{u}_A$$
$$\underline{u}2 = 0$$

Table 3.12: 0,1,2 sequence voltage-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $u0$ | p.u | Zero-Sequence-Voltage, Magnitude |
| $phiu0$ | deg | Zero-Sequence-Voltage, Angle |
| $u1r$ | p.u | Positive-Sequence-Voltage, Real Part |
| $u1i$ | p.u | Positive-Sequence-Voltage, Imaginary Part |
| $u1$ | p.u | Positive-Sequence-Voltage, Magnitude |
| $U1$ | kV | Line-Ground Positive-Sequence-Voltage, Magnitude |
| $U1l$ | kV | Line-Line Positive-Sequence-Voltage, Magnitude |
| $phiu1$ | deg | Positive-Sequence-Voltage, Angle |
| $u2$ | p.u | Negative-Sequence-Voltage, Magnitude |
| $phiu2$ | deg | Negative-Sequence-Voltage, Angle |

The variables from Table 3.12 are calculated as follows:

- $u0$ is obtained from the zero sequence complex voltage as:

$$u0 = |\underline{u}_0| \cdot \frac{uknom}{U_{nom_el}}$$

  where $uknom$ is the nominal voltage of the connected terminal and $U_{nom_el}$ is the nominal voltage of the element.

- $phiu0$ is obtained as:

$$phiu0 = \arctan\left(\frac{\Im(\underline{u}_0)}{\Re(\underline{u}_0)}\right) \cdot \frac{180}{\pi}$$

- $u1r$ is obtained from the positive sequence complex voltage as:

$$u1r = \Re(\underline{u}_1) \cdot \frac{uknom}{U_{nom_el}}$$

- $u1i$ is obtained from the positive sequence complex voltage as:

$$u1i = \Im(\underline{u}_1) \cdot \frac{uknom}{U_{nom_el}}$$

- $u1$ is obtained from the positive sequence complex voltage as:

$$u1 = |\underline{u}_1| \cdot \frac{uknom}{U_{nom_el}}$$

- $U1$ is obtained as:
  for AC elements:

$$U1 = u1 \cdot U_{nom_el}/\sqrt{3}$$

  for AC/BI elements:

$$U1 = u1 \cdot U_{nom_el}/2$$

- $U1l$ is obtained as:
  for 2-phase and 3-phase terminals:

$$U1l = u1 \cdot U_{nom_el}$$

  for 1-phase terminals:

$$U1l = U1$$

- $phiu1$ is obtained as:

$$phiu1 = \arctan\left(\frac{\Im(\underline{u}_1)}{\Re(\underline{u}_1)}\right) \cdot \frac{180}{\pi}$$

- $u2$ is obtained from the negative sequence complex voltage as:

$$u2 = |\underline{u}_2| \cdot \frac{uknom}{U_{nom_el}}$$

- $phiu2$ is obtained as:

$$phiu2 = \arctan\left(\frac{\Im(\underline{u}_2)}{\Re(\underline{u}_2)}\right) \cdot \frac{180}{\pi}$$

### 3.2.5 0,1,2 sequence current-related variables for elements

In addition to the phase quantities, quantities in the positive, negative and zero sequence are available as well. $\underline{I}_1$, $\underline{I}_2$ and $\underline{I}_0$ are obtained when the phase values are transformed to symmetrical components.

For 3-phase elements:

$$\begin{aligned}\underline{i}0 &= \tfrac{1}{3}\left(\underline{i}_A + \underline{i}_B + \underline{i}_C\right)\\ \underline{i}1 &= \tfrac{1}{3}\left(\underline{i}_A + a\cdot\underline{i}_B + a^2\cdot\underline{i}_C\right)\\ \underline{i}2 &= \tfrac{1}{3}\left(\underline{i}_A + a^2\cdot\underline{i}_B + a\cdot\underline{i}_C\right)\end{aligned}$$

where $a = \angle 120^\circ$

For BI-phase elements $(180^\circ)$:

$$\begin{aligned}\underline{i}0 &= \tfrac{1}{2}\left(\underline{i}_A + \underline{i}_B\right)\\ \underline{i}1 &= \tfrac{1}{2}\left(\underline{i}_A - \underline{i}_B\right)\\ \underline{i}2 &= 0\end{aligned}$$

For 2-phase elements $(120^\circ)$:

$$\begin{aligned}\underline{i}0 &= \tfrac{1}{\sqrt{3}}\left(\underline{i}_A + \underline{i}_B\right)\\ \underline{i}1 &= \tfrac{1}{\sqrt{3}}\left(\underline{i}_A - \underline{i}_B\right)\\ \underline{i}2 &= 0\end{aligned}$$

For 1-phase elements:

$$\begin{aligned}\underline{i}0 &= 0\\ \underline{i}1 &= \underline{i}_A\\ \underline{i}2 &= 0\end{aligned}$$

Table 3.13: 0,1,2 sequence current-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $I0$ | kA | Zero-Sequence Current, Magnitude |
| $I0{\times}3$ | kA | 3*I0 |
| $phii0$ | deg | Zero-Sequence Current, Angle |
| $i0$ | p.u | Zero-Sequence Current, Magnitude |
| $i0r$ | p.u | Zero-Sequence Current, Real Part |
| $i0i$ | p.u | Zero-Sequence Current, Imaginary Part |
| $I1$ | kA | Positive-Sequence Current, Magnitude |
| $phii1$ | deg | Positive-Sequence Current, Angle |
| $i1$ | p.u | Positive-Sequence Current, Magnitude |
| $i1r$ | p.u | Positive-Sequence Current, Real Part |
| $i1i$ | p.u | Positive-Sequence Current, Imaginary Part |
| $I2$ | kA | Negative-Sequence Current, Magnitude |
| $phii2$ | deg | Negative-Sequence Current, Angle |
| $i2$ | p.u | Negative-Sequence Current, Magnitude |
| $i2r$ | p.u | Negative-Sequence Current, Real Part |
| $i2i$ | p.u | Negative-Sequence Current, Imaginary Part |
| $i1P$ | p.u. | Positive-Sequence Active Current |
| $i1Q$ | p.u. | Positive-Sequence Reactive Current |
| $I1P$ | kA | Positive-Sequence Active Current |
| $I1Q$ | kA | Positive-Sequence Reactive Current |
| $i2P$ | p.u. | Negative-Sequence Active Current |
| $i2Q$ | p.u. | Negative-Sequence Reactive Current |
| $I2P$ | kA | Negative-Sequence Active Current |
| $I2Q$ | kA | Negative-Sequence Reactive Current |
| $phiu0i0$ | deg | Angle between Voltage and Current in zero sequence system |
| $phiu1i1$ | deg | Angle between Voltage and Current in positive sequence system |
| $phiu2i2$ | deg | Angle between Voltage and Current in negative sequence system |
| $ubfacI$ | % | Current Unbalance Factor |

The result variables from Table 3.13 are calculated as follows:

- $I0$ is obtained as the magnitude of the zero sequence current:

$$I0 = |\underline{I}_0|$$

- $I0{\times}3$ is obtained as $3 \cdot I0$ for three phase, as $2 \cdot I0$ for two phase and as $I0$ for single phase systems.
- $phii0$ is obtained as:

$$phii0 = \arctan\left(\frac{\Im(\underline{I}_0)}{\Re(\underline{I}_0)}\right) \cdot \frac{180}{\pi}$$

- $i0$ is obtained as:

$$i0 = \frac{I0}{I_{nom_el}}$$

  where $I_{nom_el}$ is the nominal current of the element.

- $i0r$ is obtained as:

$$i0r = \frac{\Re(\underline{I}_0)}{I_{nom_el}}$$

- $i0i$ is obtained as:

$$i0i = \frac{\Im(\underline{I}_0)}{I_{nom_el}}$$

- $I1$ is obtained as the magnitude of the positive sequence current:

$$I1 = |\underline{I}_1|$$

- $phii1$ is obtained as:

$$phii1 = \arctan\left(\frac{\Im(\underline{I}_1)}{\Re(\underline{I}_1)}\right) \cdot \frac{180}{\pi}$$

- $i1$ is obtained as:

$$i1 = \frac{I1}{I_{nom_el}}$$

where $I_{nom_el}$ is the nominal current of the element.

- $i1r$ is obtained as:

$$i1r = \frac{\Re(\underline{I}_1)}{I_{nom_el}}$$

- $i1i$ is obtained as:

$$i1i = \frac{\Im(\underline{I}_1)}{I_{nom_el}}$$

- $I2$ is obtained as the magnitude of the negative sequence current:

$$I2 = |\underline{I}_2|$$

- $phii2$ is obtained as:

$$phii2 = \arctan\left(\frac{\Im(\underline{I}_2)}{\Re(\underline{I}_2)}\right) \cdot \frac{180}{\pi}$$

- $i2$ is obtained as:

$$i2 = \frac{I2}{I_{nom_el}}$$

where $I_{nom_el}$ is the nominal current of the element.

- $i2r$ is obtained as:

$$i2r = \frac{\Re(\underline{I}_2)}{I_{nom_el}}$$

- $i2i$ is obtained as:

$$i2i = \frac{\Im(\underline{I}_2)}{I_{nom_el}}$$

- $I1P$, $I1Q$, $i1P$ and $i1Q$ are obtained as:

$$I1P = I1 \cdot \cos(\varphi_1)$$
$$I1Q = I1 \cdot \sin(\varphi_1)$$
$$i1P = i1 \cdot \cos(\varphi_1) = \frac{I1P}{I_{nom_el}}$$
$$i1Q = i1 \cdot \sin(\varphi_1) = \frac{I1Q}{I_{nom_el}}$$

where $\varphi_1$ is the angle of the complex apparent power in the positive sequence, corresponds to the angle $phiu1i1$.

- $I2P$, $I2Q$, $i2P$ and $i2Q$ are obtained as:

$$I2P = I2 \cdot \cos(\varphi_2)$$
$$I2Q = I2 \cdot \sin(\varphi_2)$$
$$i2P = i2 \cdot \cos(\varphi_2) = \frac{I2P}{I_{nom_el}}$$
$$i2Q = i2 \cdot \sin(\varphi_2) = \frac{I2Q}{I_{nom_el}}$$

  where $\varphi_2$ is the angle the complex apparent power in the negative sequence, corresponds to the angle $phiu2i2$.

- $phiu0i0$ is obtained as:

$$phiu0i0 = phiu0 - phii0$$

- $phiu1i1$ is obtained as:

$$phiu1i1 = phiu1 - phii1$$

- $phiu2i2$ is obtained as:

$$phiu2i2 = phiu2 - phii2$$

- $ubfacI$ is obtained as:

$$ubfacI = \frac{i2}{i1} \cdot 100$$

  If number of phases (without neutral) is less then three, it is set to zero.

### 3.2.6 0,1,2 sequence power-related variables for elements

The total complex apparent power is the sum of apparent powers of all phases:

$$\underline{S}_{sum} = \underline{S}_A + \underline{S}_B + \underline{S}_C + \underline{S}_N$$

where:

$$\underline{S}_A = \underline{U}_A \cdot \underline{I}_A^*$$
$$\underline{S}_B = \underline{U}_B \cdot \underline{I}_B^*$$
$$\underline{S}_C = \underline{U}_C \cdot \underline{I}_C^*$$
$$\underline{S}_N = \underline{U}_N \cdot \underline{I}_N^*$$

and the sequence powers:

For 3-phase elements:

$$\underline{S}_1 = 3 \cdot \underline{U1} \cdot \underline{I1}^*$$
$$\underline{S}_2 = 3 \cdot \underline{U2} \cdot \underline{I2}^*$$
$$\underline{S}_0 = 3 \cdot \underline{U0} \cdot \underline{I0}^*$$

For AC/BI elements ($180°$):

$$\underline{S}_1 = 2 \cdot \underline{U1} \cdot \underline{I1}^*$$
$$\underline{S}_2 = 0$$
$$\underline{S}_0 = 2 \cdot \underline{U0} \cdot \underline{I0}^*$$

For 2-phase elements ($120°$):

$$\underline{S}_1 = 3/2 \cdot \underline{U1} \cdot \underline{I1}^*$$
$$\underline{S}_2 = 0$$
$$\underline{S}_0 = 3/2 \cdot \underline{U0} \cdot \underline{I0}^*$$

where

$$\underline{S}_{sum} = \underline{S}_1 + \underline{S}_2 + \underline{S}_0$$

Table 3.14: 0,1,2 sequence power-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $Psum$ | MW | Total Active Power |
| $Qsum$ | Mvar | Total Reactive Power |
| $Ssum$ | MVA | Total Apparent Power |
| $cosphisum$ | | Total Power Factor |
| $tanphisum$ | | Total tan(phi) |
| $P1$ | MW | Positive Sequence Active Power |
| $Q1$ | Mvar | Positive Sequence Reactive Power |
| $P2$ | MW | Negative Sequence Active Power |
| $Q2$ | Mvar | Negative Sequence Reactive Power |
| $ubfacS$ | % | Power Unbalance Factor |

The result variables from Table 3.14 are calculated as follows:

- $Psum$ is obtained as:

$$Psum = \Re(\underline{S}_{sum})$$

- $Qsum$ is obtained as:

$$Qsum = \Im(\underline{S}_{sum})$$

- $Ssum$ is obtained as:

$$Ssum = |\underline{S}_{sum}|$$

- $cosphisum$ is obtained as:

$$cosphisum = \cos(\varphi_s)$$

  where the angle is defined as:

$$\varphi_s = \arctan\left(\frac{\Im(\underline{S}_{sum})}{\Re(\underline{S}_{sum})}\right)$$

- $tanphisum$ is obtained as:

$$tanphisum = \tan(\varphi_s)$$

- $P1$ is the positive sequence active power of the element.

$$P1 = \Re(\underline{S1})$$

- $Q1$ is the positive sequence reactive power of the element.

$$Q1 = \Im(\underline{S1})$$

- $P2$ is the negative sequence active power of the element.

$$P2 = \Re(\underline{S2})$$

- $Q2$ is the negative sequence reactive power of the element.

$$Q2 = \Im(\underline{S2})$$

- $ubfacS$ is calculated as a ratio between $\Delta_{max_abs}$ and the average of absolute apparent powers. $\Delta_{max_abs}$ is the biggest absolute difference between the apparent power of a phase and the average apparent power. In this calculation the neutral connection is not taken into account.

### 3.2.7 Miscellaneous variables (min/max values) for elements

Table 3.15: Miscellaneous variables (min/max values) for elements

| Name | Unit | Description |
|---|---|---|
| $Tfct$ | s | Fault Clearing Time |
| $Brkload$ | % | Breaker Loading |
| $Imax$ | kA | Maximum Current |
| $Smax$ | MVA | Maximum Power |

The result variables from Table 3.15 are calculated as follows:

- $Tfct$ is the minimum from the fault clearing times: $TfctPh{:}A$, $TfctPh{:}B$, $TfctPh{:}C$, $TfctPh{:}N$.
- $Brkload$ is the maximum breaker loading from: $BrkloadPh{:}A$, $BrkloadPh{:}B$, $BrkloadPh{:}C$, $BrkloadPh{:}N$.
- $Imax$ is the biggest value of the maximum currents per phase from low-voltage load flow analysis (coincidence of low-voltage loads is considered).
- $Smax$ is obtained as:

$$Smax = \sqrt{3} \cdot uknom \cdot Imax$$

## 3.3 Result variables for grouping elements

See Section 2.3 for the balanced load flow calculation.

# 4 Linear DC Load Flow calculation

For the linear DC calculation all the terminal magnitudes are set to $1p.u.$ and the losses are neglected. The calculation delivers only active power and voltage angles as results.

## 4.1 Result variables for terminals

As described in Section 1.2.2, the result variables from the *Bus Results* set from the port elements are equivalent to the result variables from the *Currents, Voltages and Currents* set for the terminals.

The result variables available for terminals after a linear DC Load Flow calculation are presented in the following sub chapters.

### 4.1.1 Voltage-related variables for terminals

Table 4.1: Voltage-related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $u$ | p.u. | Voltage, Magnitude |
| $u1$ | p.u. | Positive-Sequence Voltage, Magnitude |
| $upc$ | % | Voltage, Magnitude |
| $U$ | kV | Line-Ground Voltage, Magnitude |
| $Ul$ | kV | Line-Line Voltage, Magnitude |
| $phiu$ | deg | Voltage, Angle |
| $phiurel$ | deg | Voltage, Relative Angle |

The result variables from Table 4.1 are calculated as follows:

- $u$ has a constant value of 1 p.u..
- $u1$ has a constant value of 1 p.u..
- $upc$ has a constant value of 100 %.
- $U$ is obtained as:

$$U = 1 \cdot uknom/\sqrt{3}$$

  where $uknom$ is the nominal voltage of the terminal.

- $Ul$ is obtained as:

$$Ul = 1 \cdot uknom$$

- $phiu$ is a result from the calculation.
- $phiurel$ is obtained using the *Initial Angle of Bus Voltage* $phiini$ basic data parameter as:

$$phiurel = phiu - phiini \cdot \frac{180}{\pi}$$

### 4.1.2 Power-related variables for terminals

Table 4.2: Power-related variables for terminals

| Name | Unit | Description |
|---|---|---|
| $Pgen$ | MW | Generation, Active Power |
| $Pmot$ | MW | Motor Load, Active Power |
| $Pload$ | MW | General Load, Active Power |
| $Pcomp$ | MW | Compensation (Losses) |
| $Pnet$ | MW | External Networks, Active Power |
| $Pflow$ | MW | Power Flow, Active Power |

The result variables from Table 4.2 are calculated as follows:

- $Pgen$ is the active power sum of all synchronous, asynchronous and static generators connected to the terminal. If generation is defined in the MV Load (*ElmLodmv*), the generated active power is added to this sum.
- $Pmot$ is the active power sum of all synchronous and asynchronous motors connected to the terminal.
- $Pload$ is the active power sum of all loads connected to the terminal.
- $Pcomp$ is the active power sum of all shunts and filters connected to the terminal.
- $Pnet$ is the active power sum of all external network elements connected to the terminal.
- $Pflow$ is the sum of all active power flowing out of the terminal.

### 4.1.3 Max. power and loading for busbars

Table 4.3: Max. power and loading

| Name | Unit | Description |
|---|---|---|
| $Pmax$ | MW | Max. Active Power |
| $loading$ | % | Loading |

The maximum power results for busbars are only available for busbars with bay configuration. The topological order is defined by the position of the bays, see corresponding parameter of the bay element (*ElmBay*, parameter *position*).
For each section of the busbar is a corresponding maximum power calculated. $Pmax$ is then the maximum power of all sections.

If the busbar has a busbar type (*TypBar*) with a given rated current ($Ir$, see the Load Flow page) assigned, the loading for the busbar is calculated as follows:

$$loading = Pmax/\sqrt{3} \cdot Ir \cdot U_{nom} \cdot 100$$

where $U_{nom}$ is the nominal voltage of the busbar in kV.

## 4.2 Result variables for elements

The result variables available for for single- and multiple-port elements after a linear DC Load Flow calculation are presented in the following sub chapters.

### 4.2.1 Power-related variables for elements

Table 4.4: Power-related variables for elements

| Name | Unit | Description |
|---|---|---|
| $P$ | MW | Active Power |
| $Q$ | Mvar | Reactive Power |

The result variables from Table 4.4 are calculated as follows:

- $P$ is a calculation result.
- $Q$ is always 0 Mvar.

## 4.3 Result variables for grouping elements

### 4.3.1 Total sum results

Table 4.5: Total sum-related variables (generation and consumption)

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $TotgenP$ | MW | Total Generation, Active Power | Total P gen. |
| $TotconsP$ | MVA | Total Consumption, Active Power | Total P cons. |
| $TotP$ | MW | Total Demand | |
| $TotLoadP$ | MW | Total Load, Active Power | |
| $UnsupP$ | MW | Unsupplied Demand | |
| $SupP$ | MW | Supplied Demand | |

The “Total Generation, Active Power”, $TotgenP$, is the sum of all active power fed into the system; e.g. for generators when $P > 0$ and for loads when $P < 0$.

The “Total Consumption, Active Power”, $TotconP$, is the sum of all active power consumption from the system; e.g. for loads when $P > 0$ or generators when $P < 0$.

The demand quantities ($TotP$, $UnsupP$ and $SupP$) take all loads and all motors into account. The total demand, $TotP$, and the total load, active power, $TotLoadP$, is then:

$$TotP = LoadPnom + MotPnom$$
$$TotLoadP = LoadP + MotP$$

where $LoadPnom$ is the nominal power of all loads (refer to Table 4.9), and $MotPnom$ is the sum of the nominal active power of all motors. The active power is split into unsupplied and supplied loads and motors.
$TotLoadP$ is the actual active power of all loads ($LoadP$) (refer to Table 4.9) and the sum of the active power for all motors ($MotP$) (refer to Table 4.8).

### 4.3.2 Sum results for generators

The following elements are taken into account:

- Synchronous Generators (*ElmSym*, configured as a generator)
- Asynchronous Generators (*ElmAsm*, configured as a generator)
- Static generators (*ElmGenstat*)
- PV Systems (*ElmPvsys*)
- DC Generators (with one or two terminals) (*ElmDcm* and *ElmDcmbi*, configured as a generator)
- DFIG Generators (*ElmAsmsc*, configured as a generator)
- MV loads (*ElmLodmv*, only the generation part

Table 4.6: Sum-related variables for generators

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $GenP$ | MW | Generators, Active Power | Generators, P |
| $GenPnom$ | MW | Generators, Nominal Active Power | |
| $GenPdiff$ | MW | Generators, difference between nominal and actual active power | |
| $GenPdisp$ | MW | Generators, Dispatch Active Power | |
| $GenPmism$ | MW | Generators, difference between actual and dispatched active power | |
| $GenPspinres$ | MW | Generators, difference between maximum and actual active power | |
| $cst_disp$ | X[1]/h | Total Production Costs | |

[1]: X is replaced by the currency unit of the project (USD, EUR, GBP, ... ).

$GenP$ is the sum of the active power of all generators assigned to the grouping object, regardless of the sign; e.g. if the generator consumes power (negative active power).

The nominal active power, $GenPnom$, is calculated based on the rated apparent power and rated power factor of the generators. The difference between the nominal and actual active power, $GenPdiff$, is then $GenPdiff = GenPnom - GenP$.

The dispatched active power $GenPdisp$ is the sum of the entered dispatched active power including the wind generation scaling factor of the zone (*ElmZone*) and the generator scaling factor from the Load Flow command. The difference between the actual and dispatched active power is then $GenPmism = GenP - GenPdisp$.

The difference between the maximum and actual active power ($GenPspinres$) is only for generators with the option *Consider for region spinning reserve* enabled (see the Load Flow page ⇒ Advanced). The external grid (*ElmXnet*) is also taken into account.
Note: The option is enabled by default for synchronous machines and disabled for static generators, PV systems, asynchronous machines and external grids. DFIG machines and MV loads (*ElmAsmsc* and *ElmLodmv*) are not taken into account.

$$GenPspinres = \sum (P_{max} - P_{gen})$$

where $P_{gen}$ is the active power and $P_{max}$ the maximum active power of the generator.

The total production cost ($cst_disp$) is calculated as follow and only considers synchronous generators (*ElmSym*), static generators (*ElmGenstat*, external grids (*ElmXnet*) and asynchronous generators (*ElmAsm*) configured as a DFIG.

### 4.3.3 Sum results for external networks

The following elements are taken into account:

- External Grids (*ElmXnet*)
- AC Voltage Sources (with one or two terminals) (*ElmVac* and *ElmVacbi*), but only if not used as an earthing model ($voltage\ setpoint < 1.^{-6}$ ).
- DC Voltage Sources (with one or two terminals) (*ElmDcu* and *ElmDcubi*), but only if not used as an earthing model ($|voltage\ setpoint| < 1.^{-6}$ ).
- Batteries (with one or two terminals) (*ElmBattery* and *ElmBatterybi*)

Table 4.7: Sum-related variables for external networks

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $NetP$ | MW | External Networks, Active Power | External Networks, P |

$NetP$ is the sum of the active power of all external network models assigned to the grouping object, regardless of the sign (consumption or infeed of power).

### 4.3.4 Sum results for motors

The following elements are taken into account:

- Synchronous Motors (*ElmSym*, configured as a motor)
- Asynchronous Motors (*ElmAsm*, configured as a motor)
- DC Motors (with one or two terminals) (*ElmDcm* and *ElmDcmbi*, configured as a motor)
- DFIG Motors (*ElmAsmsc*, configured as a motor)

Table 4.8: Sum-related variables for motors

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $MotP$ | MW | Motor Loads, Active Power | Motors, P |

The variable $MotP$ is the sum of the active power of all motors assigned to the grouping object, regardless of the sign; e.g. if the motor infeeds power (negative active).

### 4.3.5 Sum results for loads

The following elements are taken into account:

- General Loads (*ElmLod*)
- MV Loads (*ElmLodmv*, load part only)
- LV Loads (*ElmLodlv*)
- DC Loads (with one or two terminals) (*ElmLoddc* and *ElmLoddcbi*)
- Trains (AC or DC) (*ElmTrain*)
- Lines with line loads (*ElmLne* with *ElmLodlvp*)

Table 4.9: Sum-related variables for loads

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $LoadP$ | MW | Loads, Active Power | Loads, P |
| $LoadPnom$ | MW | Loads, Nominal Active Power | Loads, P |
| $LoadPdiff$ | MW | Loads, difference between nominal and actual active power | Loads, dP(Un-U) |
| $LoadPdem$ | MW | Loads, Active Power Demand | |
| $LoadPmism$ | MW | Loads, difference between actual active power and demand | |

$LoadP$ is the sum of the active power of all loads assigned to the grouping object, regardless of the sign; e.g. if the load infeeds power (negative active power).

The nominal active power, $LoadPnom$, is calculated based on the dispatched active power of the loads, including the entered scaling factor (*scale0*), the load scaling factor of the corresponding zone (*ElmZone*) and the load scaling factor of the Load Flow command.

The difference between the nominal and actual active power, $LoadPdiff$, is then $LoadPdiff = LoadPnom - LoadP$.
It should be noted that the difference can be caused by, for example, the load flow being calculated using the *Distributed slack by load* option, or when the loads are voltage dependent or scaled via feeder load scaling.

The active power demand for loads ($LoadPdem$) is equal to the nominal active power of the loads. The differences between the actual active power and demand, $LoadPmism$, is then $LoadPmism = LoadP - LoadPdem$.

### 4.3.6 Interchange flow, Import and Export

Table 4.10: Interchange flow variables

| Name | Unit | Description | Short Descr. |
|---|---|---|---|
| $ExportP$ | MW | Export, Active Power | Export, P |
| $ImportP$ | MW | Import, Active Power | Import, P |
| $InterP$ | MW | Interchange Flow, Active Power | Interchange Flow, P |
| $ptot$ | MW | Total Area Exchange Power | |

For the export and import active power, $ExportP$, $ImportP$, all models are considered where the grouping is different e.g. for lines, transformers between different grids, zones or areas. Elements with only one terminal connection are taken into account when the corresponding terminal is in a different grouping to that of the element. It should be noted that the export power and import power quantities are always positive.

The interchange flow of the active power ($InterP$) is positive when the power is exported and calculated as follows:

$$InterP = ExportP - ImportP$$

The total area exchange power, $ptot$, is equal to the interchange flow active power, $InterP$, and is only available for grids (*ElmNet*).

### 4.3.7 Statistics

Table 4.11: Statistics

| Name | Description | Classes |
|---|---|---|
| $n_ele$ | Number of Elements | |
| $n_area$[1] | Number of Connected Grids | |
| $n_volt$ | Number of Voltage Levels | |
| $n_sta$[2] | No. of Substations | *ElmSubstat* and *ElmTrfstat* |
| $n_bar$ | No. of Busbars | *ElmTerm*, Usage: Busbar |
| $n_term$ | No. of Terminals | *ElmTerm*, Usage: Junction Node or Internal Node |
| $n_lne$ | No. of Lines | *ElmLne* |
| $n_2wt$[2] | No. of 2-w Trfs. | *ElmTr2* and *ElmTrb* |
| $n_3wt$[2] | No. of 3-w Trfs. | *ElmTr3* |
| $n_4wt$[2] | No. of 4-w Trfs. | *ElmTr4* |
| $n_sym$ | No. of syn. Machines | *ElmSym* |
| $n_asm$ | No. of asyn. Machines | *ElmAsm* and *ElmAsmsc* |
| $n_stg$ | No. of Static Generators | *ElmGenstat* |
| $n_pvs$ | No. of PV Systems | *ElmPvsys* |
| $n_lod$ | No. of Loads | *ElmLod*, *ElmLodmv*, *ElmLodlv*, *ElmLoddc*, *ElmLoddcbi*, *ElmIac*, *ElmIacbi*, *ElmDci*, *ElmDcibi*, *ElmTrain* |
| $n_svs$[2] | No. of SVS | *ElmSvs* |
| $n_shnt$[2] | No. of Shunts/Filters | *ElmShnt*, *ElmFilter* and *ElmVac*,*ElmDcu* used as an earthing element ($uset < 1^{-6}$) |
| $n_other$[2] | No. of Other Elements | |
| $n_cust$ | Number of Customers | *ElmLod*, *ElmLodmv*, *ElmLodlv*, *ElmLoddc*, *ElmLoddcbi*. In addition are line loads (*ElmLodlvp*) also taken into account |
| $n_cunsu$[2] | Number of unsupplied Customers | see $n_cust$ |
| $n_unsup$[2] | Number of unsupplied Loads | see $n_lod$ and in addition motors are also taken into account (*ElmSym*, *ElmAsm*, *ElmAsmsc*, *ElmDcm*, *ElmDcmbi*) |

Notes:

- [1]: For zones (*ElmZone*), a corresponding parameter is available: Number of Connected Zones, $n_zones$; and for areas, *ElmArea*: Number of Connected Areas, $n_areas$.
- [2]: Not available for zones (*ElmZone*) and areas (*ElmArea*).

In addition, the following statistics are available:

Table 4.12: Statistics

| Name | Description |
|---|---|
| $NrOvlBranch$ | Number of overloaded branch components |

where:

- $NrOvlBranch$: The number of elements with more than one connection (pure neutral wire connections are not taken into account) where $loading > 100\%$

The following are only available for grids (*ElmNet*):

Table 4.13: Overall statistics for summary grid

| Name | Description |
|---|---|
| $n_iso$ | No. of Isolated Areas |
| $n_isooff$ | No. of Unsupplied Isolated Areas |

An unsupplied isolated area is, for example, a terminal without a reference machine.